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Adversarial Machine Learning Use of Reverse Engineering for Anomaly Detection in Industrial Control Systems

Study Project Report

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1 Introduction

Industrial Control Systems (ICS) are considered to be critical systems as they are responsible for monitoring and controlling important infrastructures such as water systems, power plants, and nuclear plants. Typically, ICS consist of two major areas which are, Information Technology (IT), and Operational Technology (OT). IT focuses more on the standard computer interconnections via traditional communication protocols such as Hyper Text Transfer Protocol (HTTP), and Hyper Text Transfer Protocol Secure (HTTPS). while OT focuses more on the operational level which includes special hardware and software for monitoring and controlling sensors and actuators. OT hierarchy usually consists of field, control, and supervisory layers. Each layer is responsible for certain tasks which works as pipeline. Field layer includes sensors and actuators, while control layer consists of number of Programmable Logic Controllers (PLCs) which are responsible for managing and controlling all field layer devices. Sensors measure the current state of the underlying physical process, then they report it to PLCs. PLCs implement certain control logic which is used to evaluate the incoming sensor readings and take decisions based on them. Based on these decisions, PLCs may send certain commands to the field actuators to adjust the current state of the physical process. In the supervisory layer, there exist Supervisory Control and Data Acquisition (SCADA) system performs a high-level visualization and control over the control and field layers and provides an overview of the current process state to the working engineer (operator). OT used to be *air-gapped*, which means that they were designed to be isolated from external environments, and they were secure via *obsecurity*. Currently industrial organizations are increasingly connecting their OT network with the corporate networks to improve business and operational efficiency, but this comes at a cost. ICS are exposed to cyber-physical attacks such as security breaches in cyberspace that adversely affect the physical process. To defend ICS from such attacks, IT is important to adopt multi-layered approach. First, IT is desirable to block cyber attacks before they reach the OT network. Indeed, most attacks against ICS start by gaining access to the business network and then laterally moving to the OT network. Second, IT is important to have anomaly detection tools that works as outlier detection, which are responsible for identification and observation of events that deviate from the normal behavior, and report such cases to the working operators to handle them. Anomaly detection has a long history in the field of statistics, where analysts and scientists would study charts looking for any element that appeared abnormal. Today, anomaly detection leverages artificial intelligence and machine learning to automatically identify unexpected changes in a data set's normal behavior. Generative adversarial networks (GANs) are machine learning framework where two neural networks - a generator and a discriminator - compete against each other in a zero-sum game to produce realistic, synthetic data. The generator creates fake data, while the discriminator evaluates it against real data, and the main goal is to teach the discriminator how it can differentiate normal data from malicious data. Major problem in ICS is that some of the used protocols are vendor specific and not open-source, which prevents many researchers and developers from building anomaly detection tools which are

able to understand the protocol and allow defenders to detect attacks. One solution to bypass such problem is to develop reverse engineering methods in a trial to understand the structure of the protocol, and determine the role of each chunk. The goal of this study project is to implement and evaluate different approaches that are able to extract expressive patterns, i.e., features, from network traffic collected in the OT area of a given ICS via a reverse engineering approach, and to build different ML, DL, and GANs models to work as classifiers and to answer our main research question, which is: Can anomaly detection methods be designed to operate without relying on protocol knowledge and achieve good performance in detecting anomalies on reverse engineered traffic? We mainly use precision, recall and F1-score as our main metrics to evaluate the performance. We assume that the ICS operator utilizes proprietary binary protocols, and the approaches to be developed can not directly parse the content of the transmitted network packets. We also rely on performing reverse engineering method to extract the physical reading out from the network packets. Then we rely on different ML and DL methods to train a classifier with the extracted chunks in order to create a classification model. Finally this model is to be used as a anomaly detection tool to identify possible abnormal operation in ICS. In our analysis we focus on Siemens S7 communication (S7comm) protocol which is one of the leading protocols used in industrial networks. Siemens S7 PLCs are estimated to consititue over 30% of the world-wide PLC market. The main reason that we choose S7comm is its proprietary nature and it was not publicly available before it was reverse engineered. So we used 2017QUTS7Comm dataset to perform all our analysis.

2 Background

In this chapter, we present the fundamental concepts and technologies relevant to this work. These background elements provide the necessary context for understanding the datasets, preprocessing pipeline, and anomaly detection approaches presented in the subsequent chapters.

2.1 Machine Learning Terminology

2.1.1 Machine learning

Machine learning is an evolving branch of computational algorithms that are designed to emulate human intelligence by learning from the surrounding environment. They are considered the working horse in the new era of the so-called big data. Techniques based on machine learning have been applied successfully in diverse fields ranging from pattern recognition, computer vision, spacecraft engineering, finance, entertainment, and computational biology to biomedical and medical applications. These methods can help us in creating anomaly detection systems which are able to detect outliers, or classify new network packets as malicious. Usually machine learning is split into two major methods which are *Supervised learning* and *Unsupervised learning*. In machine learning we usually consider set of objects O which is divided into previously seen data S , and previously unseen data U , and the goal is to build a mathematical model M - which can be seen as mapping function which converts the input into certain output - that can learn from S , and apply its knowledge on U with generalization to get the best results.

2.1.2 Overfitting

This is a phenomenon happens when the model performs very good during the training phase, but the performance drops significantly during testing phase. Usually this means that the model can not generalize to previously unseen data, and it just memorizes the training data. We will sometimes refer to this concept when we obtain very high results on our metrics, and we try to generally avoid it.

2.1.3 Underfitting

This is a phenomenon happens when the model performs poorly during both training and testing phases. Usually this means that the underlying function is much more complex than

the ability of the model to learn. This usually means that the model has high bias and that the underlying mathematical function is more complex than our model.

2.1.4 Supervised learning

It is a method from machine learning, where we already have previously seen data S , with their corresponding labels, and we try to teach our model M based on these previous labels, in order to be able to predict the labels of U correctly. Examples are classification and regression.

2.1.5 Unsupervised learning

This is another method from machine learning, where there are no labels for the training data, instead the model try to learn patterns and relations between the data. Examples are outlier detection and clustering.

2.1.6 Distance functions

Usually when we are working in any feature space, we need to measure the distance between two points in this space. In order to do so, we need to define a distance function that can evaluate these distances. There are some famous distance functions such as the Manhattan (L_1) distance function, which computes the distance as the summation of the absolute difference between each feature for two different points. It can be seen as follows: for $p, q \in \mathbb{R}^n$, $\|dist(p, q)\|_1 = \sum_{i=1}^n |p_i - q_i|$. Another famous distance function is the Euclidean distance (L_2), which computes the square root of the sum of the squared differences between feature values. It can be seen as follows: for $p, q \in \mathbb{R}^n$, $\|dist(p, q)\|_2 = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$.

2.1.7 Levels of measurements for features

We can categorize data types into three major types. Firstly, *Nominal* data which are categorical, where we only ascertain whether value exist or not i.e. boolean, and there is no order for such data, and we can not also define distance function for them. Secondly, ordinal data where there exist an order relation (better/worse) between categories, but no uniform distance i.e. University grades (A+, B-, etc...). Finally, metric which can both differences and ratios between the values are meaningful. The values can be discrete or continuous. i.e. weight (continuous), age classes (discrete). In our evaluation, we will use the latter level. Specifically, we will use precision, recall, and F1-score to evaluate our models' performance.

2.1.8 K-Folding

The dataset is divided into k equally sized folds. In each iteration, one fold is used as the validation (test) split, while the remaining $k - 1$ folds form the training split. This process is repeated k times such that each fold serves once as validation data. Final performance metrics are obtained by averaging over all k runs.

2.1.9 F1-Score

In binary classification, the performance of a model can be evaluated using the metrics precision, recall, and F1-score. Let TP denote the number of true positives, FP the number of false positives, and FN the number of false negatives.

A true positive occurs when the model correctly predicts a positive instance:

$$TP = |\{x \in D \mid y(x) = 1 \wedge \hat{y}(x) = 1\}|$$

A false positive occurs when the model incorrectly predicts a positive instance:

$$FP = |\{x \in D \mid y(x) = 0 \wedge \hat{y}(x) = 1\}|$$

A false negative occurs when the model incorrectly predicts a negative instance:

$$FN = |\{x \in D \mid y(x) = 1 \wedge \hat{y}(x) = 0\}|$$

Precision measures the proportion of correctly predicted positive samples among all samples predicted as positive:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2.1)$$

Recall measures the proportion of correctly detected positive samples among all actual positive samples:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2.2)$$

The F1-score combines precision and recall into a single metric using their harmonic mean:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (2.3)$$

The F1-score takes values in the interval $[0, 1]$, where 1 indicates perfect precision and recall, and 0 indicates the worst possible performance. Due to the harmonic mean, the F1-score penalizes large imbalances between precision and recall. This makes it particularly suitable for anomaly detection in ICS, where datasets are often imbalanced and both false positives and false negatives are costly.

2.1.10 Activation functions

As we know, artificial neural network is high-order mathematical model that is composed from interconnected nodes or artificial neurons. Activation functions are mathematical functions that a neuron executes. So, it calculates the output of a neuron as per its inputs and the associated weights on individual input. It can be linear, such as unit function $f(x) = x$, however linear functions are not helpful in detecting complex features, because usually features are non-linear. Therefore, we tend to use non-linear activation functions. Famous activation functions are *Sigmoid*

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

But it suffers from vanishing gradient problem. Another family of non-linear activation functions are Rectified linear unit (ReLU) family, which tends to be much faster than Sigmoid, and they are usually used in the hidden layers of the network.

$$\text{ReLU}(x) = \max(0, x)$$

Here it treats all negative values as zeros. However this sometimes cause a problem which is called *dead neurons*, when the neuron's value stuck to zero always, and this leads to zero gradients as well. To solve this problem, another non-linear function was introduced which is *LeakyReLU*.

$$\text{LeakyReLU}(x) = \begin{cases} \alpha x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

It solves the problem of ReLU by introducing a very small slope in the negative region, controlled by the parameter α , where $\alpha > 0$ is a small constant (e.g., $\alpha = 0.01$). This allows small gradients to flow for negative inputs and helps avoid *dead neurons*.

2.2 Detection Methods

2.2.1 Traditional Machine Learning Detection Methods

In this work, we employ different classification approaches for detecting anomalous network traffic in ICS. These approaches can be divided into one-class and binary classification methods. While one-class methods are trained only on normal data and aim to detect deviations from learned behavior, binary are trained on labeled data to distinguish between benign and malicious traffic.

2.2.1.1 Detection Methods for One-Class Classification

One-Class Classification focuses on modeling benign data in order to detect abnormal instances as deviations from this distribution. During training, only benign samples are used. At test time, samples that lie outside the learned representation of normal behavior are classified as

anomalies. One Class Classifier (OCC) methods are particularly useful in scenarios where attack data is limited or intentionally excluded during training.

2.2.1.1.1 One Class-Support vector machine (OCSVM) The One-Class Support Vector Machine learns a decision boundary in feature space that encloses the majority of the normal training samples. The algorithm constructs a hyperplane that separates normal data from low-density regions while maximizing the margin. Samples that fall outside the learned boundary during testing are considered anomalous. [1]

2.2.1.1.2 Elliptic Envelope The Elliptic Envelope method assumes that normal data follows a multivariate Gaussian distribution. It estimates the mean vector and covariance matrix of the training data and defines an ellipsoidal confidence region around the center of the distribution. Instances that lie outside this region are classified as outliers. [9]

2.2.1.1.3 Local outlier factor (LOF) The Local Outlier Factor evaluates the local density of a data point with respect to its k nearest neighbors. A sample is considered anomalous if its local density is significantly lower than that of its surrounding data points. LOF is particularly suitable for datasets with varying density structures, as it focuses on local rather than global deviations. [1]

2.2.1.2 Detection Methods for Binary Classification

Binary classification models are trained using labeled benign and attack samples. The objective is to learn a decision function that separates the two classes in feature space.

2.2.1.2.1 Binary Support vector machine (BSVM) A binary Support Vector Machine constructs an optimal separating hyperplane between two classes by maximizing the margin between the closest samples of both classes, known as support vectors. In our setting, the BSVM distinguishes between benign and malicious network traffic. [4]

2.2.1.2.2 Random Forest Random Forest is an ensemble learning method that builds multiple decision trees using random subsets of training samples and features. Each tree produces a classification decision, and the final prediction is obtained by majority voting. Random Forest models are robust to noise, handle high-dimensional feature spaces effectively, and reduce overfitting compared to single decision trees. [4]

2.2.1.2.3 k Nearest Neighbor (KNN) The KNN algorithm is a distance-based classification method. It assigns a class label to a test instance based on the majority label among its k closest training samples in feature space. The method does not explicitly learn model parameters during training, but relies on the stored training data to make predictions. [4]

2.2.1.2.4 Ensemble Classification Ensemble classification refers to the combination of multiple independent classifiers to produce a single, consolidated prediction. Instead of relying on one model alone, several base learners are used in parallel, and their outputs are aggregated to determine the final decision. [4]

The underlying rationale is that different models may focus on different patterns within the data and therefore exhibit distinct error characteristics. By combining their predictions, it is possible to mitigate individual model weaknesses and achieve more stable performance.

2.2.2 DL Detection Methods

2.2.2.1 CNN

CNN is a class of deep neural network designed to process data with a grid-like topology, such as images or sequential signals. They employ convolutional layers that apply learnable filters across the input to detect local patterns and produce feature maps. Nonlinear activation functions and pooling operations are typically used to introduce nonlinearity and reduce dimensionality while preserving relevant information. By stacking multiple convolutional layers, CNNs can learn hierarchical feature representations that capture increasingly complex structures in the data. [3] Although originally developed for image processing, CNNs can also be applied to sequential data such as network traffic, where one-dimensional convolutions allow the model to identify local dependencies in raw packet byte sequences and learn discriminative features automatically. [2]

2.2.2.2 Residual Network (ResNet)

ResNet is a deep neural network architecture designed to enable the effective training of very deep models by introducing residual (skip) connections between layers[5]. Traditional deep neural networks often suffer from vanishing or degrading gradients as the number of layers increases, making optimization difficult and sometimes leading to reduced performance when the network becomes deeper. ResNet addresses this issue by reformulating the learning objective: instead of directly learning a desired mapping $H(x)$ from input x , each residual block learns a residual function $F(x) = H(x) - x$, and the final output of the block is computed as $H(x) = F(x) + x$. This identity-based skip connection allows gradients to propagate more easily through the network during backpropagation, thereby stabilizing training and enabling the construction of substantially deeper architectures. A typical ResNet architecture consists of multiple stages, each containing several residual blocks composed of convolutional layers, batch normalization, and non-linear activation functions such as ReLU.

2.2.3 Other Anomaly Detection Methods

2.2.3.1 GANs

GANs are a class of deep learning models designed for generative modeling through an adversarial training process. A GAN consists of two neural networks, called the generator and the discriminator, which are trained simultaneously in a competitive manner. As shown in figure 1 the generator aims to produce synthetic data samples that resemble real data by transforming random noise vectors into structured outputs, while the discriminator acts as a binary classifier that attempts to distinguish between real data samples and those generated by the generator. During training, the generator improves its ability to produce realistic samples by minimizing its failure rate against the discriminator, whereas the discriminator improves its classification capability by learning to better differentiate real from fake samples. This adversarial process can be interpreted as a minimax game, where the generator seeks to minimize the discriminator's success and the discriminator seeks to maximize it. Mathematically, this interaction is formulated as an optimization problem in which both networks update their parameters iteratively using gradient-based methods. Over time, if training converges properly, the generator learns the underlying data distribution and is able to produce highly realistic samples.

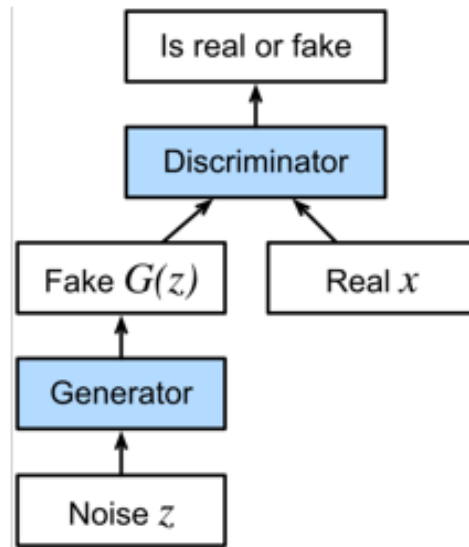


Figure 1: GAN General Architecture

2.2.3.2 Stack Denoising Autoencoder (SDA)

SDA is a deep neural network architecture designed for unsupervised feature learning and robust data representation[11]. It is constructed by stacking multiple denoising autoencoders

on top of each other, where each autoencoder layer is trained to reconstruct its input from a corrupted version of it. The key idea behind an SDA is to intentionally introduce noise into the input data and force the network to learn a compressed representation that captures the underlying structure of the data rather than memorizing it. Each individual denoising autoencoder consists of an encoder function that maps the input vector to a latent representation and a decoder function that attempts to reconstruct the original clean input. After training one layer, its encoded output is used as the input to the next layer, allowing the network to progressively learn higher-level and more abstract features. This layer-wise training strategy improves convergence and helps mitigate issues such as vanishing gradients in deeper architectures. In anomaly detection scenarios, such as in ICS, a SDA is typically trained only on benign (normal) data so that it learns a faithful representation of normal system behavior.

2.3 Provided Datasets

For our project we were presented with two different datasets. Electra¹ and 2017QUTS7Comm² The Electra S7Comm dataset and the 2017QUTS7Comm dataset differ substantially in origin, structure, and format. Electra S7Comm was published by the University of Murcia (Spain) and was collected from realistic industrial testbeds [10]. In contrast, 2017QUTS7Comm was generated in a controlled laboratory environment by the Queensland University of Technology [8].

2.3.1 Statistical Characteristics

The Electra S7Comm dataset comprises approximately 387 million packets, with roughly two thirds benign traffic and one third attack traffic. All packets belong to the S7Comm protocol over Transmission Control Protocol (TCP). Communication is limited to four host pairs, consisting of two benign communication pairs and two attack-related pairs.

The 2017QUTS7Comm dataset contains approximately 10.8 million packets and exhibits a strong class imbalance toward attack traffic. While S7Comm constitutes the majority of traffic, additional application-layer protocols such as Network Time Protocol (NTP), DHCPv6, NetBIOS Name Service (NBNS), and Multicast Domain Name Service (mDNS) are present. Both TCP and User Datagram Protocol (UDP) transport protocols occur in the dataset.

In total, 26 distinct host communication pairs are observed in 2017QUTS7Comm. However, the attack traffic is generated from a limited set of attacker endpoints, with attack activity primarily originating from a single source IP address interacting with different targets.

¹http://perception.inf.um.es/ICS-datasets/csv/electra_s7comm.zip

²https://github.com/qut-infosec/2017QUT_S7comm

3 Our Approach

In this chapter, we present our approach for processing the dataset and performing reverse engineering of the network packets. These steps are essential to transform raw proprietary ICS traffic into a structured representation that can be used for anomaly detection without relying on protocol specifications. We further describe the construction of the different Reverse-Engineered (RE) datasets. This process is particular important, as it allows us to isolate and emphasize the physical readings that are most likely related to the underlying industrial process, thereby reducing the noise introduced by protocol-related bytes. Finally, we outline the implementation of the various detection methods, which are subsequently used as anomaly detection models. The comparison of these methods enables us to evaluate how effectively the extracted representations support the identification of abnormal behavior in ICS environments.¹

3.1 Procedure Overview

Given that our research objective involves the investigation of reverse-engineering approaches, access to raw network packets was required. As the Electra dataset does not provide such data, the QUTS7Comm dataset was selected for all subsequent experiments.

The following section presents an overview of the complete processing and evaluation pipeline employed in this study project.

Figure 2 presents the complete processing pipeline of our approach. The workflow begins with the provided Packet Capture (pcap) files from the 2017QUTS7comm dataset. These packet captures are transformed and preprocessed using two alternative strategies.

The first strategy operates directly on the raw packet data (*RAW* representation). The second strategy applies a reverse-engineering procedure to extract a dynamic payload region, which is assumed to contain process-related information. After extraction, consecutive payload segments are concatenated according to a parameter p , which controls the number of packets aggregated into a single sample.

As a result, four dataset representations are generated:

- **RAW**: processed packet data without reverse engineering,
- **RE5**, **RE10**, and **RE15**: reverse-engineered representations using $p \in \{5, 10, 15\}$, respectively.

¹You can find all implementations in our repository on github: https://github.com/abdelazizSalah/Study_Project.

3 Our Approach

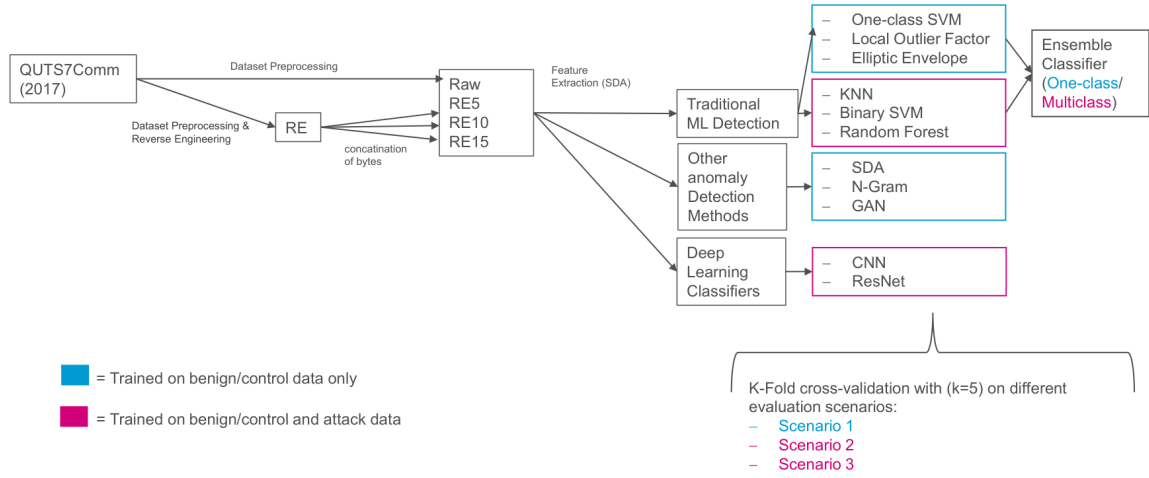


Figure 2: Overview of proposed processing and evaluation pipeline

Next, a SDA is used to extract a compact latent feature vector from each representation. These feature vectors serve as input for the traditional ML and selected DL models.

The detection problem is considered from two perspectives: outlier detection and supervised classification. Accordingly, both one-class and supervised classifiers are evaluated.

We employ six traditional ML models:

- **One-class models:**
 - One-Class Support vector machine (SVM)
 - LOF
 - Elliptic Envelope (EE)
- **Supervised models:**
 - Binary SVM
 - KNN
 - Random Forest

In addition, two DL classifiers are investigated:

- CNN
- ResNet

Finally, three additional anomaly detection approaches are evaluated:

- Reconstruction-error-based SDA
- N-gram model

- GANs-based detection

All detection methods are evaluated under three predefined evaluation scenarios using k -fold cross-validation.

3.2 Dataset Pre-processing

During the preprocessing we transform the provided PCAP files into structured datasets suitable for training and evaluating the detection methods. The process consists of label extraction, metadata parsing, and the construction of two different dataset representations: RAW and RE.

Attack labels were derived from the dataset folder structure. Each PCAP file corresponds to either benign traffic or a specific attack type, which allows assigning ground-truth labels to all contained packets. For every packet, we extracted relevant metadata including timestamps, source and destination Internet Protocol (IP) addresses, and source and destination port numbers. This metadata was used for filtering and dataset organization but was not directly provided as input to the detection models.

For the RAW representation, all packets contained in the PCAP files were included without protocol-level filtering. This includes packets without application-layer payload as well as packets unrelated to the S7 communication protocol.

Each packet was represented by its complete raw byte sequence, including Ethernet, IP, and TCP/UDP headers followed by the application-layer payload (if present). No additional parsing or content truncation was performed in this representation.

For the RE representation, only S7Comm-related application-layer data was considered. Packets were filtered according to the following criteria:

- The packet contains a TCP layer,
- Either the source or destination port equals 102 (standard S7Comm port),
- The packet length satisfies the minimum structural size of TPKT (4 bytes), COTP (3 bytes), and S7 header (12 bytes).

After identifying valid S7Comm packets, only the application-layer payload was retained. Based on the keyword position determined during the reverse engineering phase, the byte sequence was truncated such that only the payload starting from the identified keyword was used. The keyword is identified in the Reverse Engineering process, as we describe in the following section 3.3. This representation therefore focuses on protocol-relevant content rather than the full packet structure.

3.2.1 Concatenation of Bytes to create RE5,RE10,RE15

After identifying the index of the most suitable candidate keyword, we focused on extracting only the physical reading from the packets. However, we observed that extracting the physical reading from each packet individually results in very short byte sequences, which may not provide sufficient information for effective analysis.

To address this limitation, we introduced a parameter denoted as p , which defines the number of consecutive packets to be combined together. As illustrated in figure 3, we concatenate the physical readings - which are located directly after the index of the best candidate - of every p consecutive raw packets to construct a single sample in the RE dataset.

More specifically, the physical readings extracted from raw packets $p_1, p_2, \dots, p_p$ are concatenated to form single RE sample. This process is repeated sequentially over the entire raw dataset. If the total number of raw packets is not divisible by p , the remaining packets at the end of the dataset are concatenated together without further padding or duplication. This concatenation strategy enables us to achieve three main objectives. First, it allows us to focus exclusively on the physical readings as features, thereby eliminating irrelevant protocol-related bytes from the analysis. Second, by combining consecutive packets, we partially reconstruct the contextual and temporal relationship between physical readings, which may improve the representational quality of the data. Finally, it allows us to systematically evaluate the impact of varying the parameter p on the performance of the different detection methods.

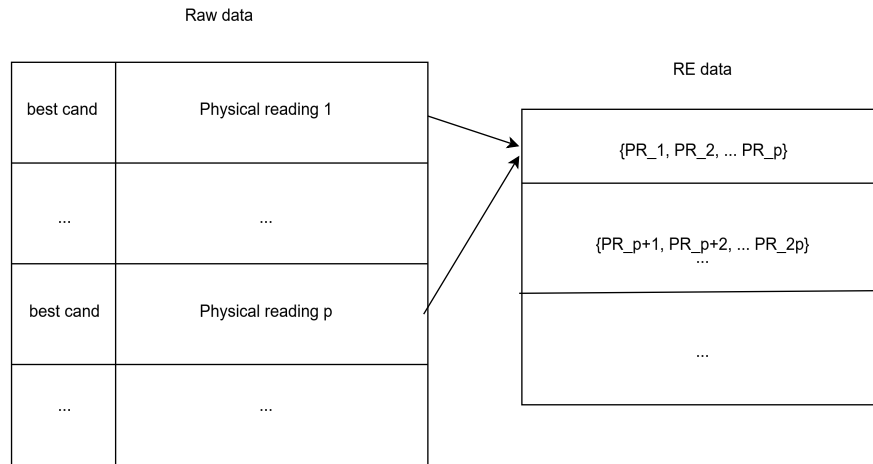


Figure 3: Concatenation of Bytes

3.3 Reverse engineering

The objective of the reverse engineering process is to identify the byte position (keyword) after which physical process readings are most likely contained within the S7 communication

payload. Since the protocol specification is not explicitly parsed, the goal is to infer structural patterns directly from the observed traffic.

3.3.1 Identifying Keyword Candidates

Firstly we identify all possible keyword candidates, by applying the following processing steps on the network traffic packets from the raw pcap files.

Session Extraction

The network traffic is divided into communication sessions. Sessions are identified based on the tuple (source IP, destination IP, source port, destination port) combined with a time window derived from packet inter-arrival times. This grouping separates independent communication instances and also distinguishes between client-to-server and server-to-client directions.

Progressive Sequence Alignment

For each session and communication direction, progressive sequence alignment is performed over all communication sessions. The purpose of this alignment is to identify recurring structural patterns across multiple messages.

Identification of Static and Dynamic Fields

Based on the aligned sequences, byte positions are classified as *static* if they remain constant across messages and as *dynamic* if their values vary.

Static byte positions typically correspond to protocol headers, control fields, or structural identifiers, while dynamic positions are potential candidates for variable data such as physical readings. Adjacent static byte positions are merged into static fields to form coherent structural blocks.

Clustering of Dynamic Fields

After separating static and dynamic fields, each dynamic field is considered as a potential keyword candidate. For each candidate, the subsequent payload region (i.e., the remaining frame behind the candidate position) is extracted and grouped into value-based clusters. The clustering is performed by grouping aligned messages according to the byte values of the candidate field: messages with identical values in the candidate interval $[start, end]$ are assigned to the same cluster.

The intuition is that if a candidate position marks the beginning of physical readings, the subsequent byte sequences should exhibit consistent value patterns across similar operational states.

3.3.2 Cluster-Similarity Analysis for Keyword Selection

After generating value-based clusters for each keyword candidate field, we analyze the resulting cluster structure to identify the best keyword candidate. For this we used the following approaches.

3.3.2.1 Payload-Length Homogeneity within Clusters

In addition to cluster-size statistics, we analyze whether messages grouped by a keyword candidate exhibit consistent payload characteristics. For each aligned message, we compute its effective payload length as the number of non-empty bytes (non-None positions) in the alignment. For each cluster with at least two messages, we compute the standard deviation of these payload lengths and aggregate the values across clusters using a size-weighted average. A low homogeneity score indicates that messages within the same cluster have similar payload lengths, which is expected when the candidate field groups messages into structurally similar types. Candidates achieving low within-cluster variability are therefore considered more likely to mark a meaningful boundary in the protocol payload.

3.3.2.2 Joint Probability

Not every candidate keyword identified during the previous steps represents a valid protocol keyword. Therefore, we introduce several constraints in order to filter out unlikely candidates.

First, we define a maximum keyword length constraint. This constraint introduces a parameter L , which represents the maximum allowed length of a keyword. Length of a keyword is defined as the number of bytes in the keyword. Any candidate keyword whose length exceeds L is discarded. This step reduces the number of potential candidates and removes fields that are unlikely to represent protocol keywords.

For the remaining candidates, we evaluate the quality of the clustering results by computing three probability measures. These probabilities assess different structural and behavioral properties of the clusters and help determine the most likely keyword candidate.

- **Message Similarity (P_m):** This probability measures the similarity between messages within and across clusters. The underlying assumption is that messages belonging to the same cluster should exhibit higher similarity than messages belonging to different clusters.

For every pair of messages, we compute a similarity score defined as the ratio between the number of identical bytes and the total number of bytes in both messages. These scores form a similarity matrix containing the similarity values for all message pairs.

Based on the clustering results for a given keyword candidate, the similarity scores are divided into two groups:

- *Intra-scores*: similarity scores between messages belonging to the same cluster.

- *Inter-scores*: similarity scores between messages belonging to different clusters.

A threshold parameter T is then introduced in order to compute two additional metrics:

- False Matching Rate (FMR):

$$FMR = \frac{\#(\text{inter-scores} > T)}{\text{total number of inter-scores}}$$

- False Non-Matching Rate (FNMR):

$$FNMR = \frac{\#(\text{intra-scores} < T)}{\text{total number of intra-scores}}$$

Using these values, the message similarity probability is defined as:

$$P_m = 1 - \frac{FMR + FNMR}{2}$$

A higher value of P_m indicates that the candidate keyword produces clusters with high internal similarity and low similarity across clusters, which suggests a good keyword candidate.

- **Remote Coupling (P_r):** This probability evaluates the relationship between clusters of client messages and clusters of server responses. The intuition behind this metric is that messages of the same type sent by a client should typically trigger similar responses from the server.

The computation proceeds as follows:

1. Client-to-server messages are clustered.
2. Server-to-client messages are clustered independently.
3. Messages are then paired according to their TCP session order.

For each client cluster we define:

- N : the number of client messages in the cluster.
- M : the largest number of corresponding server responses that belong to the same server cluster.

The remote coupling probability is then defined as:

$$P_r = \frac{M}{N}$$

A high value of P_r indicates strong correspondence between client requests and server responses, suggesting that the candidate keyword correctly separates different message types.

- **Structure Coherence (P_s):** This probability measures how consistent the internal structure of messages is within each cluster. The assumption is that messages of the same type should share a similar structural format.

For each cluster, the messages are aligned again and the number of alignment gaps introduced during this process is counted. The probability is then computed as:

$$P_s = 1 - \frac{\text{average number of gaps}}{\text{aligned message length}}$$

A higher value of P_s indicates that messages within the cluster share a consistent structure, which supports the validity of the candidate keyword.

Finally, we compute the joint probability as

$$P_t = P_m \cdot P_r \cdot P_s$$

under the assumption that the three probabilities are independent. The resulting value represents the overall likelihood that a candidate field corresponds to a valid protocol keyword. Candidates with higher values of P_t are therefore considered more likely to represent legitimate keywords. And best keyword should be the one with the highest joint probability.

Following this scheme we managed to correctly detect the best keyword in protocol S7comm, and we identified the keyword index 12 as shown in figure 4, which is *Item count field*, after which we can find the physical readings.

S7 Communication - Query								
Header						Parameter		
Protocol ID	ROSCTR	Redundancy ID	(PDU) Reference	Parameter length	Data length	Function Code (FC)	Item count	Items
32	01	00 00	01 00	00 0e	00 00	04	01	12 0a ..


 Best candidate

Figure 4: Determining best candidate index in S7comm protocol

3.4 Evaluation Strategy

To ensure a fair and systematic comparison of the implemented detection methods, a structured evaluation strategy is adopted.

3.4.1 Implementation of Evaluation Scenarios with K-Fold Cross-Validation

Three evaluation scenarios are defined to simulate different training conditions, ranging from purely benign training data to partially supervised settings with varying numbers of attack types.

- **Scenario 1:** The training set contains only benign samples.
- **Scenario 2:** The training set contains all benign samples and samples from $N - 2$ of the N attack types.
- **Scenario 3:** The training set contains all benign samples and samples from one randomly selected attack type.

In all scenarios, the test set contains benign samples as well as samples from all attack types.

All scenarios are implemented using k -fold cross-validation with 5 iterations ($k=5$).

This method is used to obtain robust performance estimates by evaluating each model on multiple train-test splits. In addition, cross-validation makes efficient use of the available data, since every sample is used for validation exactly once and for training in the remaining folds.

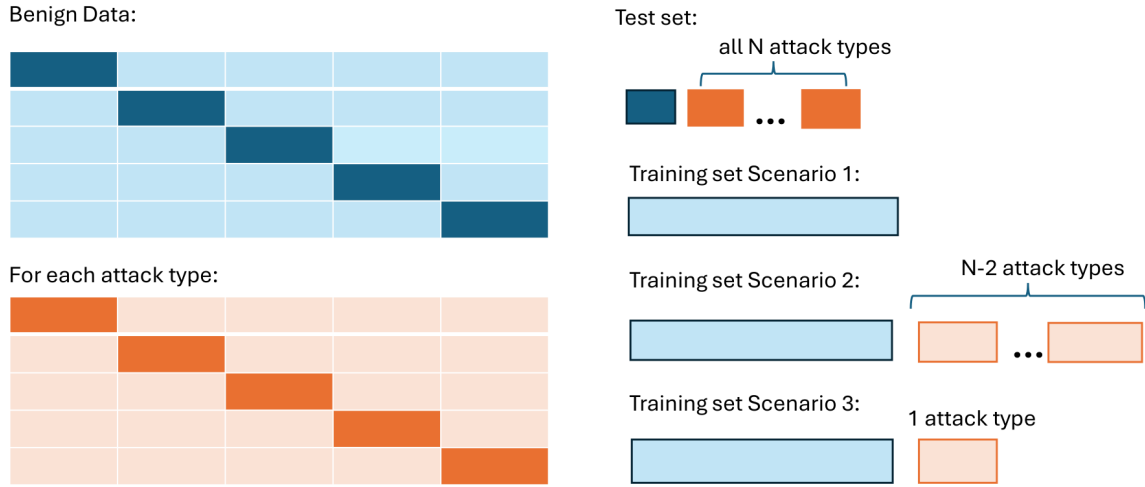


Figure 5: K-Fold cross-validation adapted to the three evaluation scenarios

To integrate the evaluation scenarios into the cross-validation procedure, the dataset is first partitioned into benign samples and separate subsets for each attack type. Cross-validation is then performed independently on each subset.

As illustrated in figure 5, in each iteration the validation fold of the benign subset is combined with the corresponding validation fold of each attack-type subset to form the test split. Consequently, the test split remains identical across all scenarios, ensuring comparability between them.

The training split differs depending on the scenario:

- In **Scenario 1**, only the $k - 1$ benign training folds are used.
- In **Scenario 2**, the benign training folds are combined with the $k - 1$ training folds of $N - 2$ attack types.
- In **Scenario 3**, the benign training folds are combined with the $k - 1$ training folds of one randomly selected attack type.

As a result, the benign portion of the training data remains identical across scenarios, while only the included attack types vary. This design ensures that performance differences between scenarios can be attributed to changes in attack diversity rather than differences in test data.

3.4.2 Evaluation Methods

For each implemented model, hyperparameter tuning is performed to determine optimal parameter configurations. It was usually performed using grid-search approach using the standard recommended values for each parameter. Subsequently, experiments are conducted using these optimized parameters within each fold and evaluation scenario. Performance metrics are averaged across all cross-validation runs.

Since the test sets in all scenarios contain both benign and attack samples and are not class-balanced, accuracy is not used as the primary evaluation metric. Instead, we report the F1-score, which provides a more informative assessment under skewed class distributions.

Beyond overall detection performance, we additionally analyze feature importance, error overlaps, and computational complexity (runtime and memory consumption) for the traditional ML classifiers. These analyses are conducted for the RAW dataset representation and for the RE15 representation to enable a detailed comparison of classifier behavior under full and reduced feature sets.

3.4.2.1 Feature Importance Analysis

To better understand the contribution of the extracted features to the classification decision, we perform a feature importance analysis for the traditional machine learning classifiers. Since different models provide different interpretability mechanisms, model-specific importance measures are used.

For the binary linear SVM, feature importance is computed as the absolute magnitude of the learned weight vector, i.e., $|\mathbf{w}|$, which reflects the contribution strength of each input feature to the decision boundary.

For the Random Forest classifier, we use the built-in impurity-based importance measure (mean decrease in impurity), which quantifies how much each feature contributes to reducing node impurity across all trees in the ensemble.

For all other traditional ML classifiers, permutation-based feature importance is applied. In this approach, the values of a single feature are randomly permuted, and the decrease in model performance is measured. A larger performance drop indicates higher feature relevance.

The resulting importance scores are normalized and ranked in descending order. This analysis allows us to identify which autoencoder-generated features contribute most to attack detection and to compare feature reliance across different models.

3.4.2.2 Error Overlap Analysis

To analyze the overlap of classification errors among different detection methods, we construct Venn diagrams for each evaluation scenario. Each circle represents the set of misclassified test samples produced by a particular classifier. Intersections between circles indicate samples that were misclassified by multiple models simultaneously.

This analysis provides insight into whether different classifiers fail on the same instances or whether their errors are complementary. A high degree of overlap suggests that the models capture similar patterns, whereas low overlap indicates complementary behavior.

3.4.2.3 Runtime and Memory Consumption

In addition to detection performance, we evaluate the computational complexity of each method in terms of runtime and memory consumption. For each model, we measure:

- Feature generation time, measured separately for (i) training the autoencoder and (ii) extracting features from the trained model
- Average training time and average inference time for each traditional ML detection method,
- Peak memory usage during feature generation, measured separately for (i) training the autoencoder and (ii) extracting features from the trained model
- Peak memory usage during training and inference for each traditional ML detection method.

Runtime is measured using system time functions, while memory consumption is monitored using profiling tools. The reported values correspond to the average across all cross-validation folds.

This analysis allows us to compare the practical feasibility of traditional machine learning methods.

3.5 Implementation of Anomaly detection methods

This subsection describes the traditional machine learning methods used for anomaly detection. First, we explain how latent features are extracted from the packet data using the SDA. These features are then used as input to our classifiers implemented in our framework.

3.5.1 Traditional ML detection methods

To apply traditional machine learning classifiers to network packet data, a fixed-length feature representation is required. Therefore, we first extract compact latent representations from the raw packet data using the SDA.

3.5.1.1 Feature Extraction using the SDA

Traditional ML classifiers such as SVM, KNN, and Random Forest require a fixed-length, low-dimensional feature representation and do not natively handle variable-length byte sequences. Therefore we employ a SDA as a feature extractor to obtain compact latent representations of the preprocessed packet data. These latent feature vectors are subsequently used as input to all traditional machine learning detection methods.

Each sample in a dataset representation (RAW, RE5, RE10, RE15) is given as a byte-vector. To ensure a fixed input dimensionality, every byte-vector is either truncated or zero-padded to a predefined length M and then normalized to the range $[0, 1]$. In our experiments, M was set to the length of the longest packet (or concatenated RE sequence) within the corresponding representation. For each dataset representation, the SDA is trained exclusively on benign samples and on the training set from the current k-fold iteration. Using only benign samples enables the model to learn a compressed representation of normal traffic patterns.

After training, the SDA is used to encode all samples of the corresponding representation. Concretely, each input byte-vector is forwarded through the encoder, and the output vector of the bottleneck layer are extracted as the latent feature vector. In our implementation, the bottleneck produces a 32-dimensional feature vector per input sample, resulting in a compact representation that is used for subsequent training and evaluation of the traditional ML classifiers.

3.5.1.2 Ensemble Classifier Implementation

In order to reduce model-specific errors and leverage complementary strengths to achieve an overall more robust and stable detection performance we combine the traditional ML classifiers into an ensemble classifier.

As illustrated in figure 6 the ensemble classifier combines the predictions of three independently trained traditional machine learning models. Each base classifier is trained separately using the corresponding training split of the cross-validation procedure. During inference, all three classifiers produce an individual prediction for each test instance.

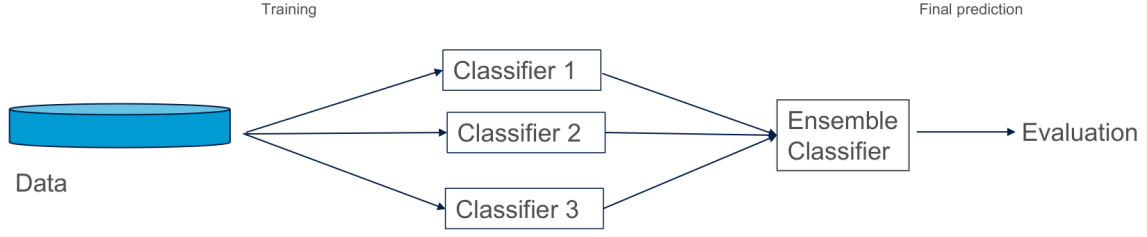


Figure 6: Ensemble Classifier Structure

The final decision of the ensemble is obtained by aggregating the three individual predictions according to a predefined aggregation strategy. In our implementation, the following prediction modes are supported:

- **Random:** One of the three predictions is selected uniformly at random.
- **Majority:** The final prediction corresponds to the class predicted by the majority of the three classifiers.
- **All:** A sample is classified as an attack only if all three classifiers predict it as an attack; otherwise, it is classified as benign.

Since the project includes six traditional ML detection methods: three operating as one-class (outlier) detectors and three operating as binary classifiers—we implemented two operational ensemble modes:

- **One-Class Mode:** The ensemble aggregates the predictions of the three outlier detectors (One-Class SVM, LOF, and Elliptic Envelope).
- **Binary Classification Mode:** The ensemble aggregates the predictions of the three binary classifiers (Binary SVM, k-Nearest Neighbors, and Random Forest).

3.5.2 DL detection methods

In addition to traditional machine learning approaches, we implement deep learning based detection methods that can learn hierarchical feature representations directly from the raw byte sequences of network packets. Unlike traditional classifiers, deep neural networks can automatically capture complex spatial patterns and dependencies in packet data without relying on manually engineered features.

In particular, we investigate CNN and ResNet. CNNs are well suited for analyzing structured sequential data such as byte sequences, as convolutional filters can learn local patterns that frequently occur in network packets. ResNet extends this idea by introducing residual connections that allow training deeper neural architectures while mitigating the vanishing gradient problem. These models enable the system to learn more expressive representations of the packet structure and improve anomaly detection performance.

3.5.2.1 CNN

We implemented a CNN to perform binary attack detection directly on the preprocessed byte sequences (i.e., without using the autoencoder-generated features). Similar to the feature generation pipeline, all inputs are converted to a fixed length of M bytes by truncation or zero-padding to ensure a consistent input shape. For our experiments, M was set to the length of the longest packet in the corresponding dataset representation. Each sample is provided to the network in the shape $(M, 1)$, where the last dimension represents the single input channel.

The CNN consists of four convolutional blocks followed by a dense classification head. Each convolutional block contains two repetitions of the sequence *Conv1D* $\rightarrow$ *Batch Normalization* $\rightarrow$ *Activation* $\rightarrow$ *Dropout*. The convolution layers learn local byte-patterns by applying trainable filters across the input sequence, batch normalization stabilizes training by normalizing intermediate activations, and dropout reduces overfitting by randomly deactivating a fraction of units during training. The number of filters increases with network depth (e.g., 8, 16, 32, 64), enabling the model to learn progressively more abstract representations.

To reduce the temporal resolution between blocks, we apply downsampling either via Max-Pooling1D after each block or by using stride-2 in the first convolution of blocks two to four (depending on the selected configuration). After the fourth block, the resulting feature maps are transformed into a fixed-size vector using a global pooling operation. By default, we apply Global Average Pooling, which aggregates each feature map by averaging over its length dimension and produces a compact representation for the dense layers.

Finally, the network applies a fully connected layer (with batch normalization, activation, and dropout) and a Softmax output layer with two neurons to predict the probability of the classes *benign* and *attack*. The model is optimized using Adam and categorical cross-entropy loss. Architecture-related hyperparameters (e.g., activation function, kernel size, dropout rate, downsampling strategy, and pooling type) as well as optimization parameters were selected via grid search on the training split.

3.5.2.2 ResNet

In the context of anomaly detection in ICS, ResNet can be adapted to process one-dimensional byte sequences of network packets using one-dimensional convolutional layers. The residual connections help preserve low-level structural information while progressively extracting higher-level representations, which improves classification performance on complex and high-dimensional network traffic data. Due to its stable training behavior and strong representational capacity, ResNet is particularly suitable for modeling subtle deviations in proprietary binary communication protocols. ResNet accepts as an input $m \times n$ complete network packets or the parsed physical readings, where m is the number of packets, and n is the number of bytes per each packet.

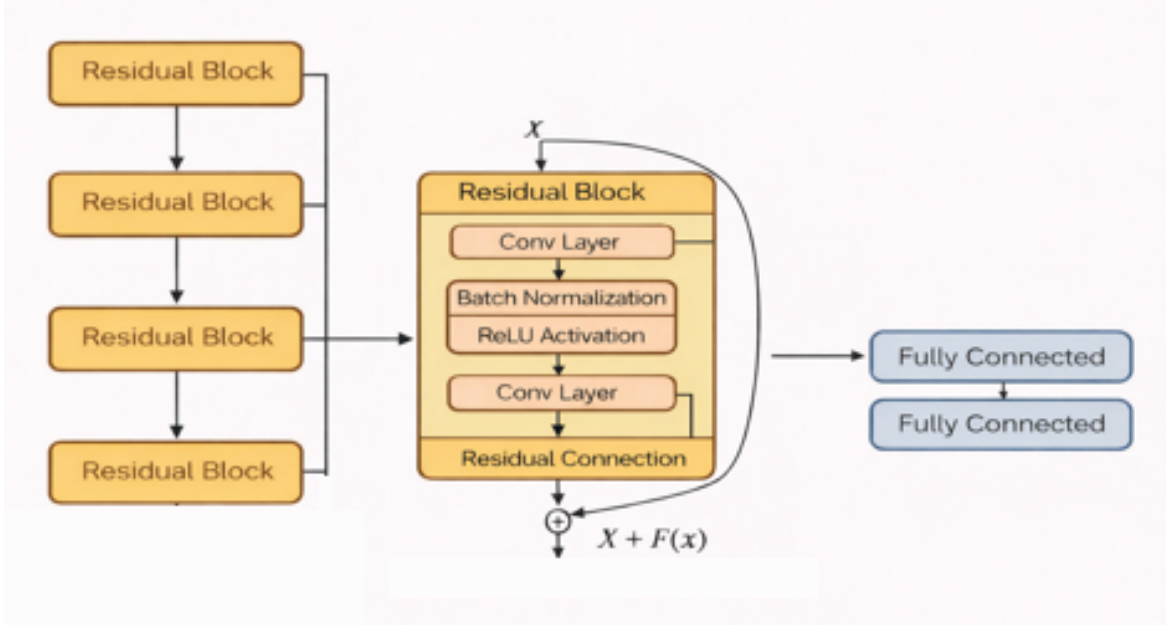


Figure 7: ResNet18 Architecture

3.5.2.2.1 Our implementation We managed to implement ResNet18 which contains of 18 layers in total. these layers are divided into four main residual blocks as shown in figure 7 and each block consists of two convolution blocks. Each convolution block contains two convolution layers with a batch normalization and ReLU non-linear activation function in between. After the 4 residual blocks, we forward the output to two fully-connected layers, that will use the reduced data to classify the network packet using sigmoid function. We used Adam optimizer to update weights, and for the loss function we used categorical cross-entropy.

3.5.2.2.2 Statistical features Along with the raw network packets, and the reverse engineered packets, we managed to extract 5 other statistical features and append them to the ResNet.

1. Total number of bytes in a given sample packet which is counted before trimming or padding this packet.
2. Number of occurrence of the most often found byte
3. Number of occurrence of the second most often found byte,
4. Byte entropy, which measures randomness of packet bytes. It is computed as

$$E(P) = - \sum_{b=0}^{255} p(b) \log_2 p(b)$$

where $p(b)$ denotes the probability of observing byte value b in the packet.

$$p(b) = \frac{n_b}{N}$$

where n_b is number of occurrences of byte b in the given packet, and N is the size of the packet.

5. Ratio of padded bytes (RPB), which tells the model how much of the input is artificial. It is computed as

$$\text{RPB} = \frac{\text{number of bytes before padding}}{\text{total number of bytes}}$$

3.5.3 Other anomaly detection methods

In addition to the traditional ML and deep learning classifiers, we evaluate several alternative approaches for anomaly detection. These include an SDA-based reconstruction error detector, an n-gram statistical model that captures byte sequence patterns in network packets, and a GANs based method capable of modeling the distribution of normal traffic. These methods provide complementary perspectives on anomaly detection by leveraging reconstruction errors, statistical sequence modeling, and generative learning.

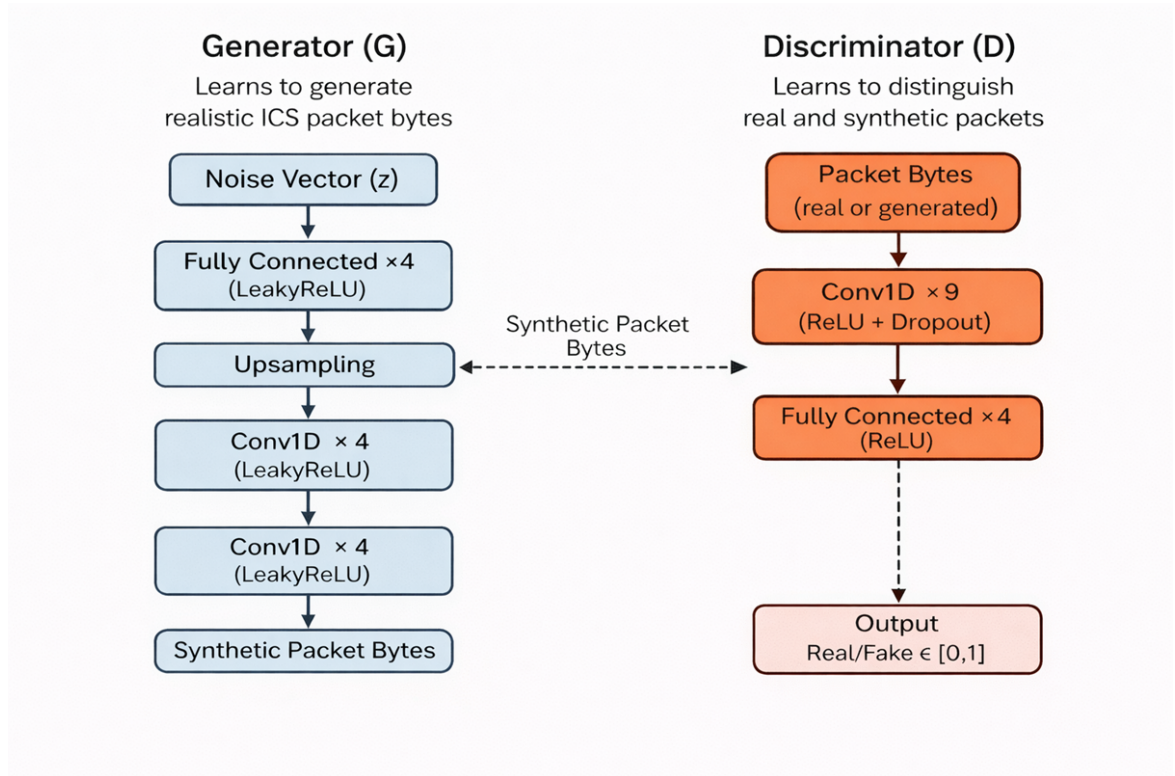


Figure 8: Discriminator and Generator architectures

3.5.3.1 SDA-Based Reconstruction Error Detection

The implemented SDA is employed as a standalone anomaly detection method based on reconstruction error. The SDA is trained exclusively on benign network traffic. During training, the model learns a compressed latent representation of normal packets and optimizes its parameters to minimize the reconstruction error between the input and the reconstructed output. Since attack packets deviate from the distribution of normal traffic, they are expected to be reconstructed less accurately, resulting in a higher reconstruction error.

For a given input byte-vector $\mathbf{x} \in \mathbb{R}^M$, the trained autoencoder produces a reconstruction $\hat{\mathbf{x}}$. The reconstruction error is computed using the Mean Squared Error (MSE):

$$\text{MSE}(\mathbf{x}) = \frac{1}{M} \sum_{i=1}^M (x_i - \hat{x}_i)^2. \quad (3.1)$$

To transform the SDA into a binary classifier, a decision threshold τ is defined based on the reconstruction error distribution of benign training samples. Concretely, we compute the per-sample MSE for all benign training packets and set τ to the 95th percentile of these errors, i.e., such that 95% of benign reconstruction errors lie below τ .

During testing, each packet is processed by the trained SDA and its reconstruction error is computed. The final classification rule is:

$$\hat{y} = \begin{cases} \text{attack}, & \text{if } \text{MSE}(\mathbf{x}) > \tau, \\ \text{benign}, & \text{otherwise.} \end{cases} \quad (3.2)$$

Packets whose reconstruction error exceeds the predefined threshold are therefore classified as anomalous.

3.5.3.2 GANs

In anomaly detection scenarios, particularly in ICS, GANs can be trained on normal data only, allowing the discriminator or generator to detect deviations from the learned distribution during testing. Because GANs are capable of learning complex, high-dimensional data distributions, they are well suited for modeling raw network packet bytes or other structured sequential inputs.

3.5.3.2.1 Discriminator Implementation Discriminator architecture as shown in figure 8 consists of CNN that consists of 9 convolution layers with ReLU activation function, along with dropout layer after all convolution layers except the first and final layers. Output is passed to four fully connected layers with ReLU activation function, and the output is generated from Sigmoid function which predicts a value in range [0,1] indicating how fake the input data is, 0 means fake data, while 1 means real data.

3.5.3.2.2 Generator Implementation Generator is reverse CNN. It uses *deconvolution* operation, which is the inverse operation of convolution, which is a method to transform noise into meaningful network packets with a certain degree of accuracy. Its architecture as shown in figure 8 consists of 4 fully connected layers with a LeakyReLU activation function, then they are followed by 8 convolution layers with LeakyReLU activation function. The first 4 layers from these 8 convolution layers are preceded by upsampling layers, the main reason for that is to increase the number of data samples in the dataset, which can help in correcting imbalances in data, enhancing the quality of generated data, and improve the model performance. We mainly used LeakyReLU here because we want to be able to regenerate the network packet out of noise input, so it can help us keeping as much neurons alive as possible to participate in the generation of the network packets. Moreover, for the generator we managed to implement custom loss function, which extract features from real and fake samples, and compute the mean feature vectors, then it computes L2 loss between mean feature vectors.

```
def feature_matching_loss(D, x_real, x_fake):  
    '''  
    Input:  
        D: Discriminator model  
        x_real: real data batch  
        x_fake: fake data batch  
    Output:  
        loss: feature matching loss value  
    '''  
    f_real = D.extract_features(x_real)  
    f_fake = D.extract_features(x_fake)  
  
    mean_real = f_real.mean(dim=0)  
    mean_fake = f_fake.mean(dim=0)  
  
    loss = torch.mean((mean_real - mean_fake) ** 2) / mean_real.numel()  
    return loss
```

3.5.3.3 N-Gram

It is a contiguous sequence of N items extracted from a larger sequence of data, where the items may represent characters, bytes, or tokens depending on the application domain. In the context of network traffic analysis, N-grams are typically constructed from raw packet bytes, meaning that each N-gram represents a sequence of N consecutive bytes within a packet. N-gram is usually defined by 2 main parameters, the value of n , which defines the number of consecutive bytes, and *stride* which define the jumping value. For example a 2-gram (bigram) with $stride = 1$ captures pairs of adjacent bytes, while a 5-gram with $stride = 1$ captures sequences of five consecutive bytes. In figure 9, we demonstrate the idea of n-grams but on string of characters, when $n=3$, and $stride=1$, we can see that we consider the first three consecutive characters *HEL* as one feature, then we increase the pointer position by the stride value which is equal to 1 in our case to next starting position, and it will be in our case letter *E*, then we take 3 consecutive characters again to form *ELL*, and we keep this process till the

end of the string. On the second example you can see the stride value is 2, so it will have the same behavior, except that when we go to the next starting index, instead of increasing the starting position by 1, we will increase it by 2, so in the second iteration instead of starting from index 1, we will start from 2 which contains letter *L* instead of letter *E*. The main idea behind N-gram modeling is that local sequential patterns contain meaningful structural information about the underlying data distribution. During the training phase, all distinct N-grams observed in benign traffic are collected and their frequencies are recorded, often using memory-efficient data structures such as **Bloom filters**. At testing time, a new packet is evaluated by measuring how many of its N-grams were previously observed during training and how frequently they occurred. We define a score for whether this packet is normal or malicious as:

$$\text{Score} = \frac{\sum_i w_i \cdot N_{\text{seen}_i}}{T}$$

Score $\in [0, 1]$, where N_{seen_i} indicates each n-gram in the packet that was seen during training, w_i represents its normalized weight how often this n-gram was seen during training, and T represents the total number of n-grams in the current packet. Packets that contain a high proportion of unseen or rarely occurring N-grams are more likely to deviate from normal behavior and can therefore be classified as anomalous, in other words, packets with low scores should be classified as malicious. N-gram-based methods are particularly suitable for anomaly detection in ICS because proprietary binary protocols often lack publicly available specifications, making direct parsing difficult. By operating directly on raw byte sequences without requiring protocol awareness, N-grams provide a simple yet effective statistical modeling technique for detecting deviations in structured network traffic.

3.5.3.3.1 Bloom Filter is a probabilistic data structure designed for efficient set membership testing with minimal memory consumption. It allows one to determine whether an element is possibly in a set or definitely not in a set. A Bloom filter consists of a fixed-size bit array and multiple independent hash functions. When inserting an element into the filter, the element is processed by each hash function, and the corresponding bit positions in the array are set to one. To check whether an element is present, the same hash functions are applied, and the bits at the resulting positions are inspected. If at least one of these bits is zero, the element is guaranteed not to be in the set. If all bits are one, the element is assumed to be in the set; however, this may result in a false positive due to hash collisions. Importantly, Bloom filters do not produce false negatives, meaning that an inserted element will never be reported as absent. The trade-off between memory usage and false positive probability depends on the size of the bit array and the number of hash functions used. In the context of N-gram-based anomaly detection for ICS, Bloom filters are particularly useful for storing large numbers of observed N-grams in a space-efficient manner, enabling fast membership queries during testing without requiring storage of the full set of N-grams.

3.5.3.3.2 Hash Functions and Double Hashing Strategy Instead of computing k independent hash functions, we apply a double hashing strategy. Two base hash values $h_1(x)$ and $h_2(x)$ are computed, and the i -th hash value is generated as:

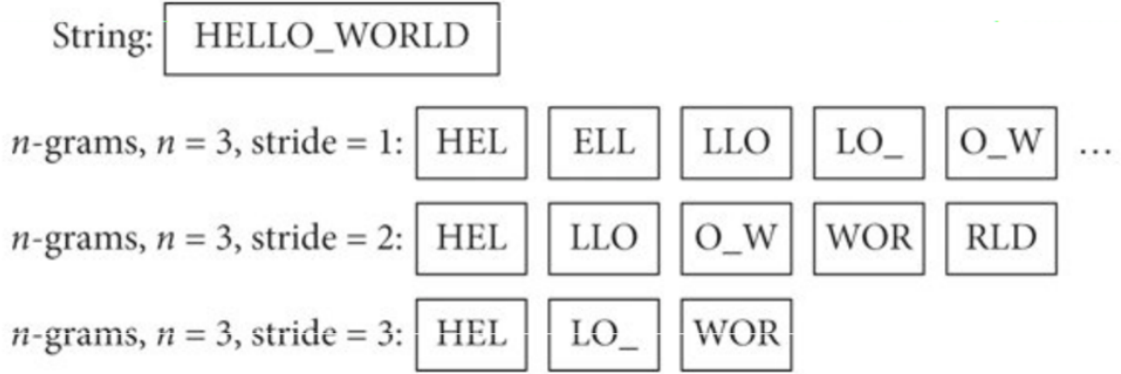


Figure 9: N-Gram Example

$$h_i(x) = (h_1(x) + i \cdot h_2(x)) \bmod m$$

where m denotes the size of the Bloom filter in bits.

This approach significantly reduces computational cost while maintaining good distribution properties and minimizing correlations between generated hash values. For h_1 and h_2 we used both a non-cryptographic and a cryptographic hash function to ensure a good balance between speed and randomness, which are MurmurHash3 and BLAKE2b respectively.

MurmurHash3 is a non-cryptographic hash function optimized for high speed and low computational overhead. It produces a uniform 64-bit output and is widely used in Bloom filters and hash tables due to its strong avalanche effect and low collision probability in practical applications.

BLAKE2b is a cryptographic hash function designed to provide both high security and high performance. It generates outputs with strong entropy and collision resistance. In our implementation, we use an 8-byte (64-bit) digest to obtain a compact yet well-distributed hash value.

```
def _hashes(self, data: bytes):
    """
    Generate k hash values for the given data using double hashing.
    Parameters:
        data: byte sequence to be hashed

    Yields:
        k hash values in the range [0, m-1]
    """

    # First hash: MurmurHash3 (fast, non-cryptographic)
    h1, _ = mmh3.hash64(data, seed=0, signed=False)
```

```
# Second hash: BLAKE2b (cryptographic, strong entropy)
blake = hashlib.blake2b(data, digest_size=8).digest()
h2 = int.from_bytes(blake, "big") | 1 # ensure h2 is odd

# Generate k hash values using double hashing
for i in range(self.number_of_hash_functions):
    yield (h1 + i * h2) % self.required_number_of_bits
```


4 Evaluation

In this chapter, we present the experimental results of the implemented detection methods. The evaluation is primarily based on tree performance metrics: precision, recall, and F1-score. By systematically comparing these metrics across the different methods, we are able to identify their respective strengths and limitations. This comparison further allows us to determine which approach achieves the best overall performance under the considered evaluation scenarios, as well as analyze the inherent trade-offs between detection accuracy and false positive or false negative rates.

4.1 Traditional ML Detection Methods

4.1.1 F1-Scores

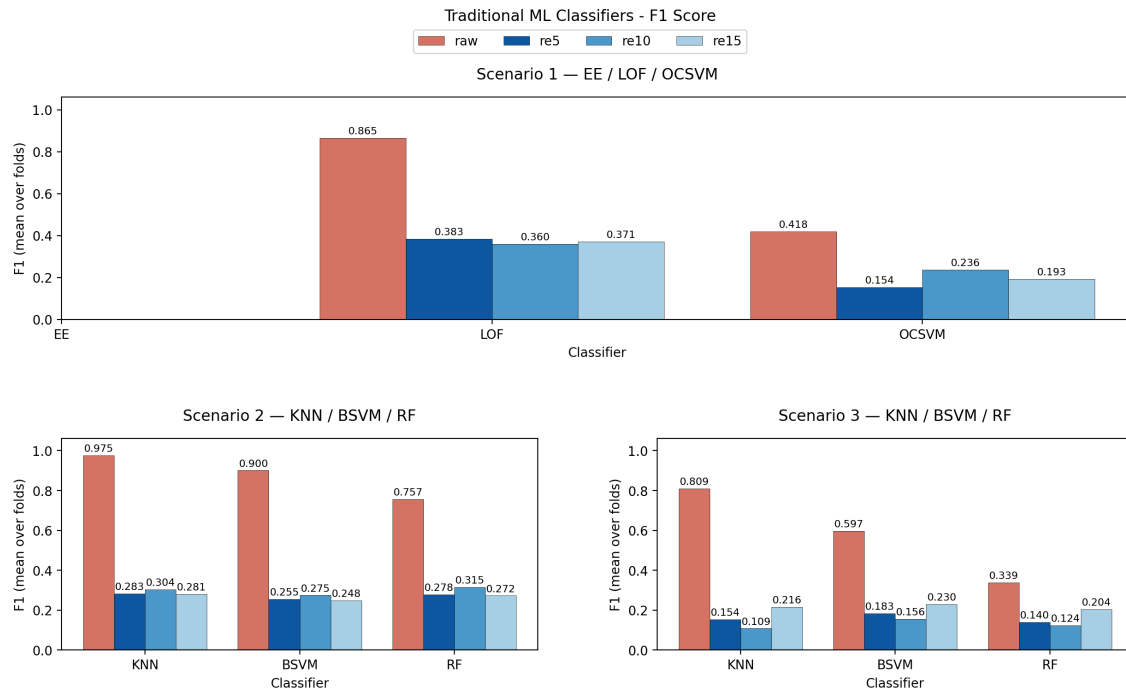


Figure 10: Traditional ML Detection Methods F1-Score

figure 10 presents the mean F1-scores of the traditional machine learning classifiers across the three evaluation scenarios. Scenario 1 evaluates the one-class methods (Elliptic Envelope, LOF,

and One-Class SVM), whereas Scenarios 2 and 3 evaluate the supervised classifiers (KNN, Binary SVM, and Random Forest). The different colors represent the dataset representations (RAW, RE5, RE10, RE15).

For the RAW representation, LOF achieves the strongest performance, clearly outperforming Elliptic Envelope and One-Class SVM. In contrast, the latter two methods achieve only moderate performance.

When using the RE representations, the performance drops substantially for all one-class methods. In particular, Elliptic Envelope fails to detect anomalies effectively, while LOF and One-Class SVM achieve only low F1-scores. This indicates that the reduced RE representations do not provide sufficient discriminative information for one-class modeling.

In Scenario 2, all supervised classifiers achieve high F1-scores when operating on the RAW representation. KNN performs best, followed closely by Binary SVM and Random Forest.

However, when switching to RE-based representations, performance decreases significantly across all classifiers. Although minor variations exist between RE5, RE10, and RE15, the results remain consistently below the RAW baseline. This suggests that the full packet structure contains important discriminative features that are partially lost in the RE representations.

Scenario 3 introduces previously unseen attack types during testing. For the RAW representation, all supervised classifiers exhibit a noticeable drop in performance compared to Scenario 2, with KNN remaining the strongest model.

Using the RE representations further degrades performance. In this setting, several classifiers approach near-random behavior.

4.1.2 Ensemble Classifier Results

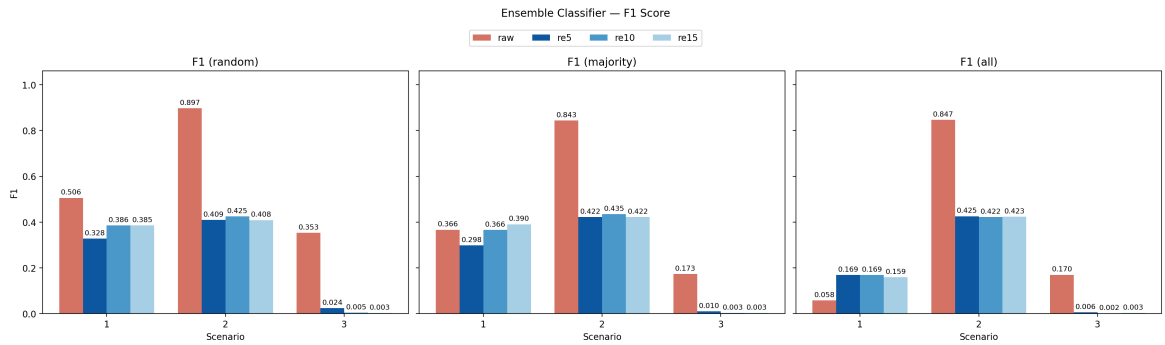


Figure 11: Ensemble Classifier F1-Score

figure 11 shows the mean F1-scores of the ensemble classifier under the three aggregation modes (random, majority, all) across all dataset representations and evaluation scenarios.

Across all representations and aggregation modes, a consistent difficulty ordering can be observed: Scenario 2 achieves the best performance, Scenario 1 yields intermediate results, and Scenario 3 consistently performs worst. This pattern is stable across RAW and all

reverse-engineered representations, indicating that the scenario characteristics have a stronger impact on performance than the chosen ensemble strategy.

In particular, Scenario 3 leads to a substantial drop in F1-score for all dataset types, reflecting the difficulty of generalizing to highly constrained or unseen attack configurations.

For the RAW representation, the random and majority strategies perform similarly and achieve strong results in Scenario 2. In contrast, the *all* strategy often leads to a noticeable performance decrease, especially in Scenario 1. This behavior suggests that at least one weak base classifier negatively influences the ensemble when unanimity is required.

Overall, majority voting provides the most stable behavior across scenarios, while the *all* strategy appears overly restrictive.

A clear performance gap is visible between RAW and all reverse-engineered representations. The RAW dataset consistently achieves substantially higher F1-scores across all scenarios and ensemble modes.

The reverse-engineered representations (RE5, RE10, RE15) show comparable behavior among each other, with RE15 generally yielding the strongest results within this group. Nevertheless, their performance remains significantly below that of the RAW data.

4.1.3 Feature Importance

To better understand which components of the 32-dimensional latent feature vector drive the classification decisions, we analyze feature importance for each traditional ML model. Each panel in the following figures corresponds to one classifier and displays the importance scores for all 32 features (f_0 – f_{31}).

Since importance definitions differ between models and are normalized independently (see 3.4.2.1), values should primarily be interpreted within each classifier rather than compared directly across models.

Positive importance values indicate that a feature contributes positively to correct classification performance, whereas negative values indicate that permuting the feature improves performance, suggesting that the feature may introduce noise or misleading signals.

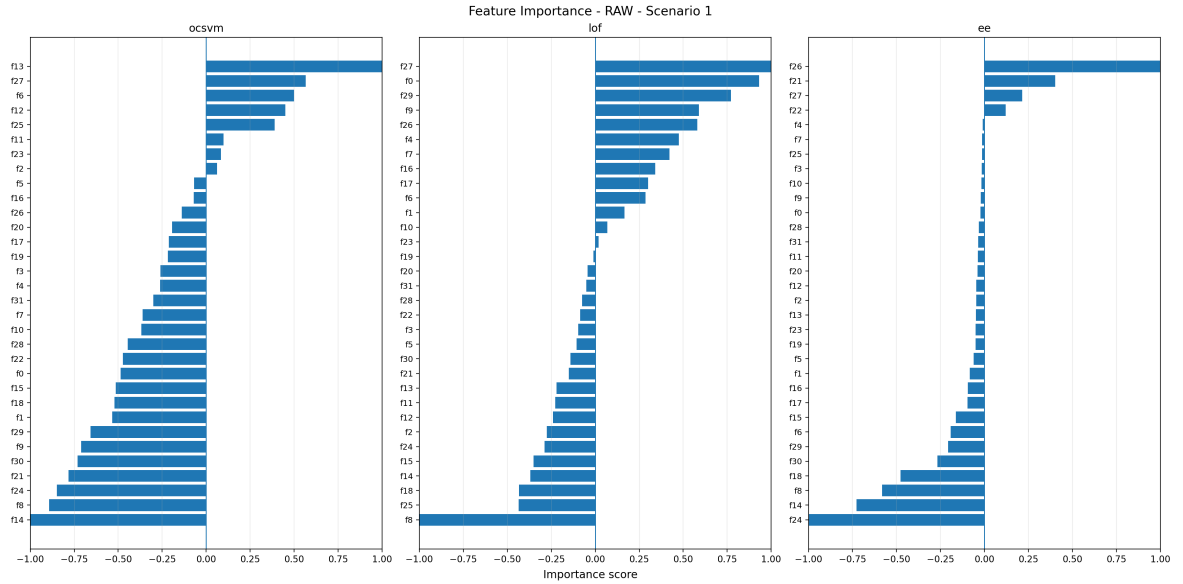


Figure 12: Feature Importance — RAW, Scenario 1

For the RAW representation in Scenario 1 (see fig. 12), several consistent patterns emerge across the one-class models. In particular, f_{27} appears among the most helpful features for all classifiers, whereas f_{14} and f_8 are consistently ranked among the most harmful.

OCSVM and EE rely strongly on a small number of dominant features (OCSVM: f_{13} ; EE: f_{26}), while LOF distributes importance more broadly across multiple components of the latent vector. Furthermore, clear sign disagreements occur between classifiers, indicating that the one-class methods extract different structural aspects from the same feature space.

In the supervised setting of Scenario 2 (fig. 13), Random Forest is dominated by a relatively small subset of features, most prominently f_{27} , followed by f_8 and f_{16} . KNN exhibits a more polarized structure with both strongly positive and strongly negative contributions, reflecting a selective reliance on specific latent dimensions.

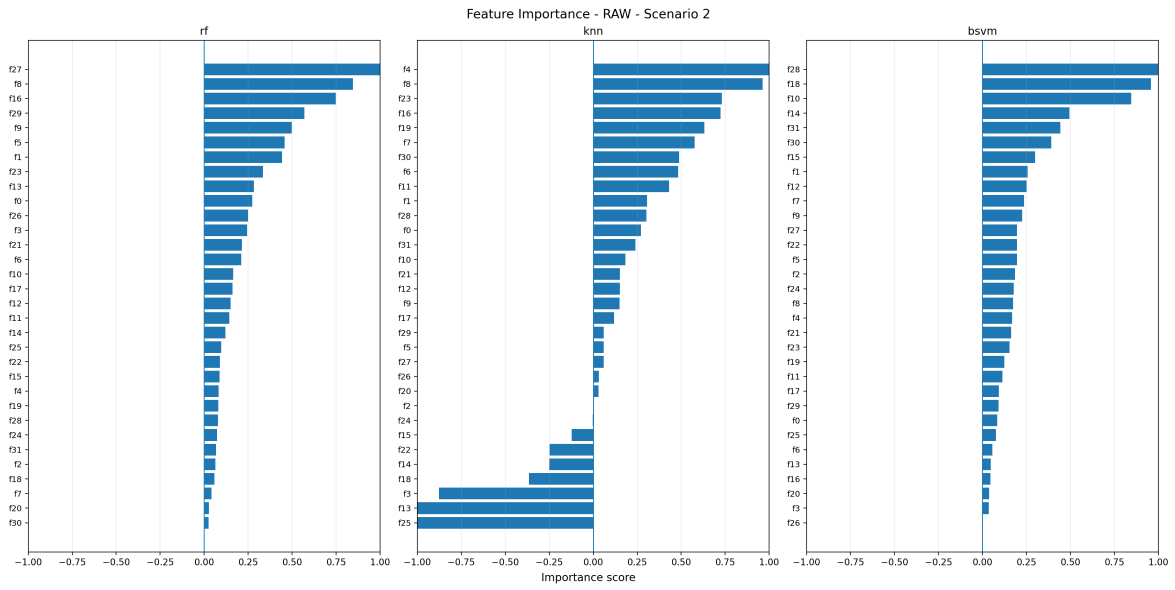


Figure 13: Feature Importance — RAW, Scenario 2

Binary SVM emphasizes a partially different subset of features, suggesting that the linear decision boundary captures directions in the latent space that differ from those exploited by tree-based or neighbor-based models.

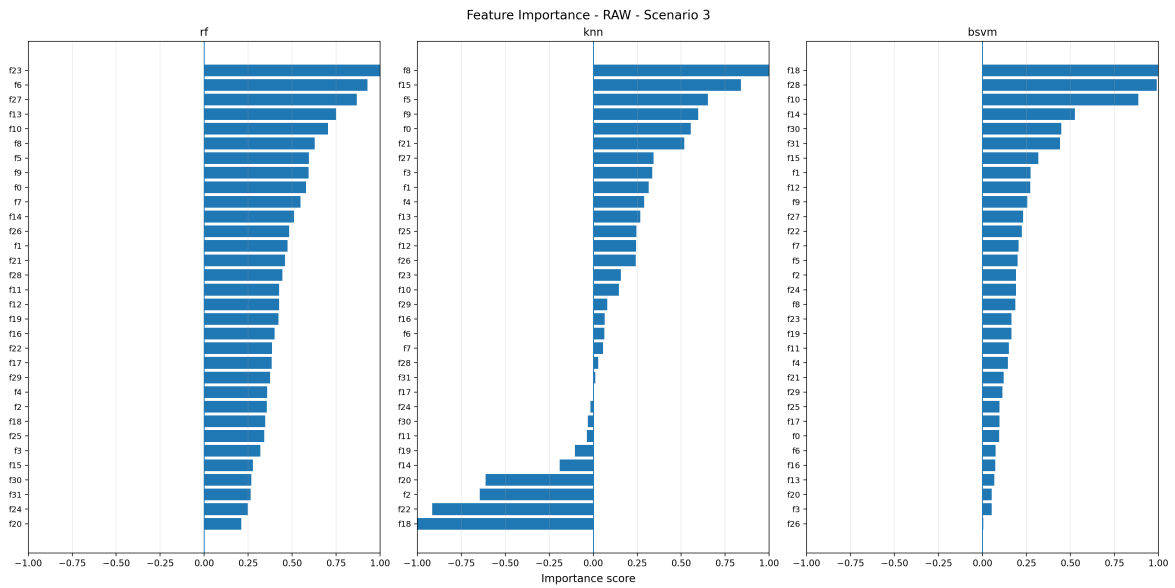


Figure 14: Feature Importance — RAW, Scenario 3

Under the more challenging generalization setting of Scenario 3 (fig. 14), importance patterns become more concentrated. Random Forest highlights a compact group of dominant features, while KNN again shows strong positive and negative contributions. Binary SVM focuses on a limited subset of high-weight components.

Notably, certain features such as f_{27} and f_8 remain consistently relevant across classifiers, indicating that specific latent dimensions retain discriminative power even when attack types differ between training and testing.

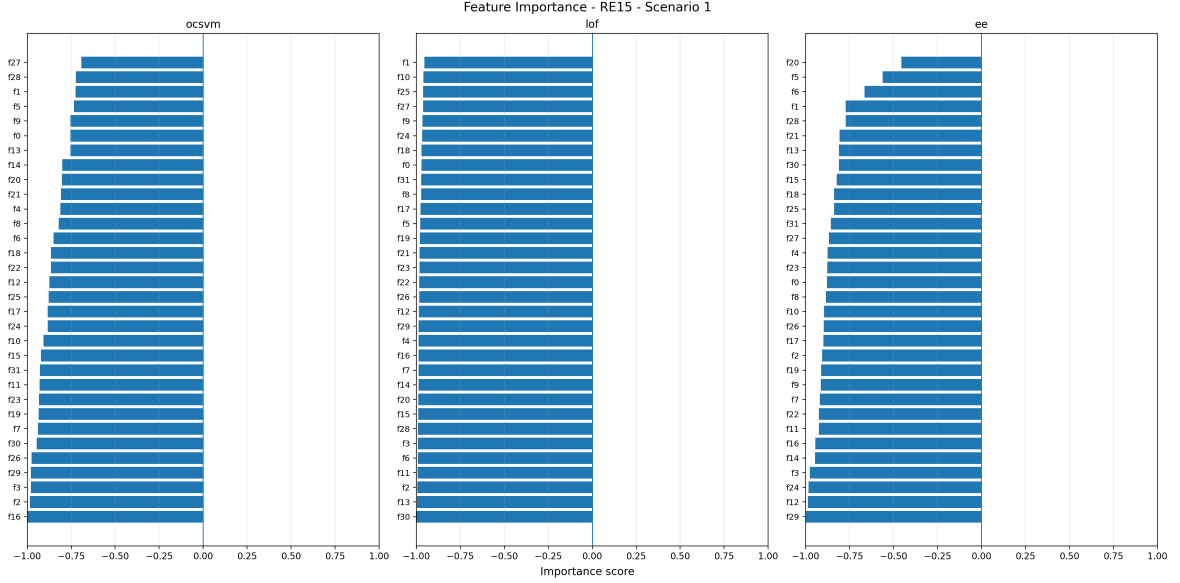


Figure 15: Feature Importance — RE15, Scenario 1

For the RE15 representation in Scenario 1 (fig. 15), the pattern differs markedly from RAW. Nearly all features receive negative importance values across classifiers, and no clearly beneficial subset emerges. This indicates that the reverse-engineered representation does not provide strong discriminative structure for one-class detection in this configuration.

In Scenario 2 with RE15 (fig. 16), the classifiers exhibit more differentiated behavior. Random Forest distributes importance relatively broadly, KNN again separates helpful from harmful features, and Binary SVM concentrates on a smaller subset of dominant weights.

Although limited cross-model overlap exists (e.g., f_{10} appears relevant for multiple classifiers), overall agreement between models is weaker than observed for the RAW representation.

In Scenario 3 under RE15 (fig. 17), the importance structure becomes considerably less distinctive. KNN assigns nearly uniform importance across features, Random Forest distributes importance broadly without clear dominance, and Binary SVM concentrates on a small subset of higher-weight components.

Overall, the RE15 representation yields less structured and less consistent importance patterns than RAW, which aligns with the observed degradation in classification performance.

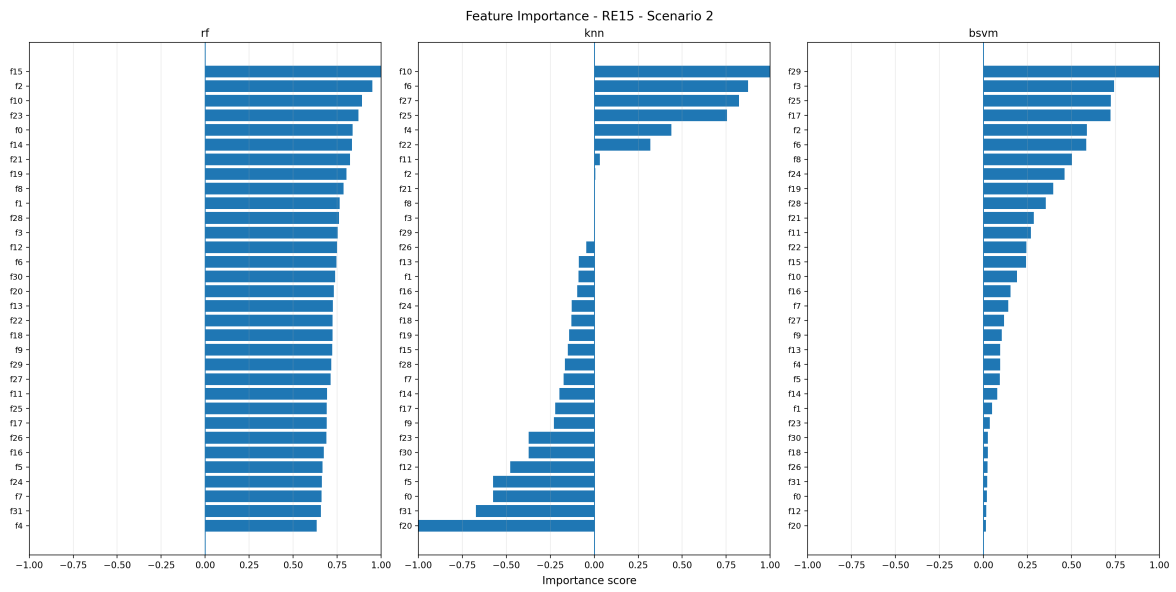


Figure 16: Feature Importance — RE15, Scenario 2

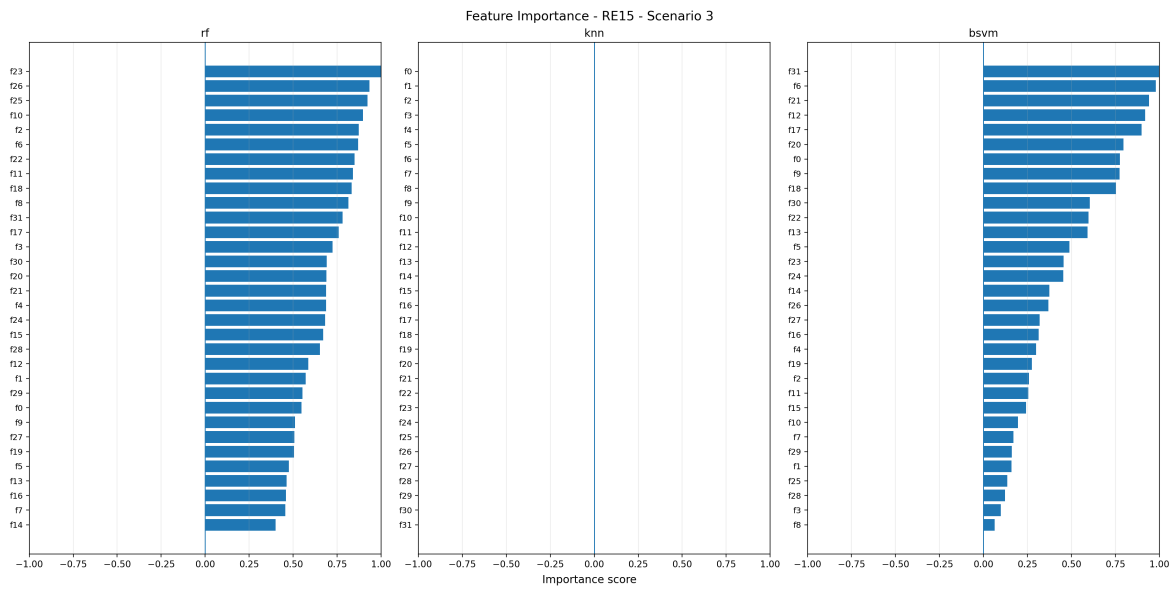


Figure 17: Feature Importance — RE15, Scenario 3

4.1.4 Venn Diagrams

The Venn diagrams visualize the overlap of misclassified samples between the three traditional ML classifiers within each scenario. Each circle represents the set of incorrectly classified instances for one classifier. Pairwise intersections indicate samples that are misclassified by two classifiers, while the triple intersection represents instances that are misclassified by all three models. Large overlap regions therefore indicate agreement in error behavior, whereas large regions without overlaps reveal classifier-specific weaknesses.

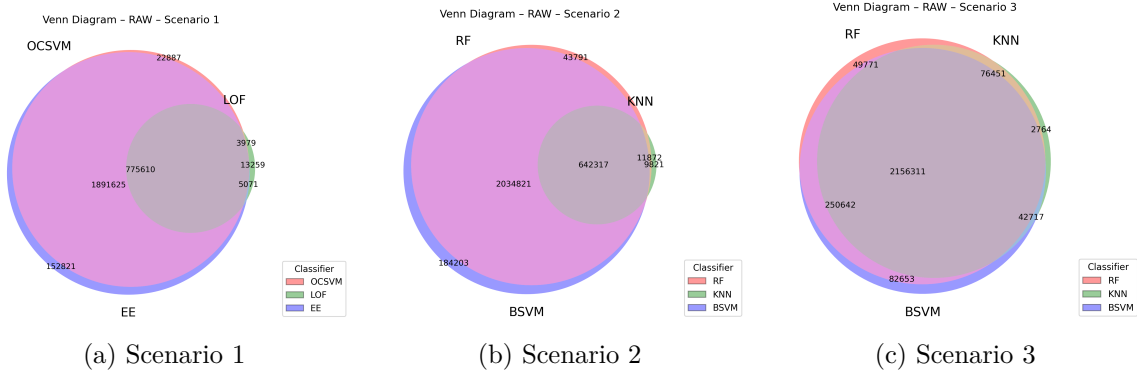


Figure 18: Error overlaps (Venn diagrams) for the RAW representation.

4.1.4.0.1 RAW Representation For the RAW dataset, the absolute number of errors is substantially higher due to the larger number of evaluated samples. As fig. 18 shows in Scenarios 1 and 2, the diagrams are dominated by large pairwise overlaps. In particular, OCSVM and EE (Scenario 1) as well as RF and Binary SVM (Scenario 2) exhibit highly similar error patterns. This suggests that these classifier pairs learn comparable decision boundaries when using RAW packet features, while the third classifier contributes only a limited amount of additional unique errors.

In Scenario 3 (RAW), the triple intersection becomes the dominant region. Most misclassified instances are shared across all classifiers, indicating that errors are not model-specific but correspond to a common subset of intrinsically difficult samples. This observation explains why ensemble aggregation provides only limited improvement in this scenario: when all base classifiers fail on the same instances, the ensemble cannot compensate.

4.1.4.0.2 RE15 Representation For the RE15 representation, the overlap structure changes across scenarios. In Scenario 1, the diagrams show considerable classifier-specific error regions (fig. 19), meaning that each model misclassifies a substantial number of samples that the others classify correctly. Although a triple overlap exists, it is not dominant, indicating divergent decision behavior.

In Scenario 2, the triple intersection becomes the largest region, while the unique error regions shrink considerably. This indicates that most misclassifications are shared among all classifiers, and only a small fraction depends on the model choice.

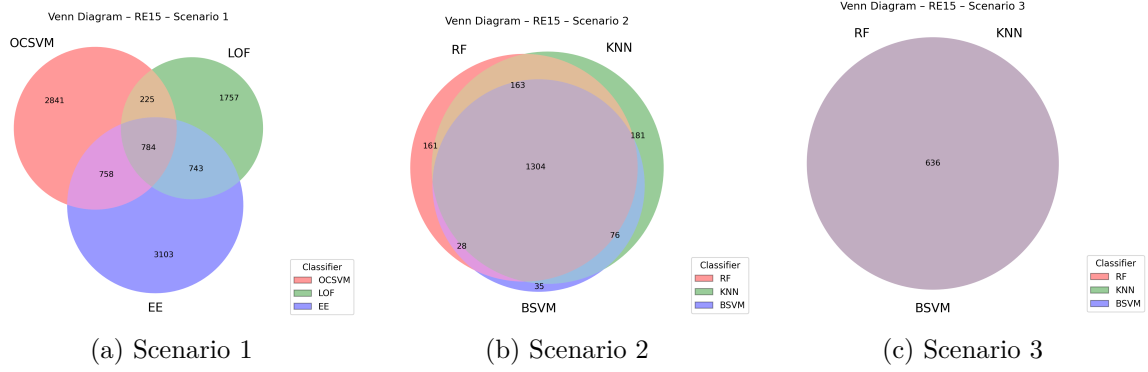


Figure 19: Error overlaps (Venn diagrams) for the RE15 representation.

In Scenario 3, the triple overlap effectively covers the entire error set. All three classifiers misclassify the same instances, representing complete agreement in failure cases. Under this condition, the ensemble classifier cannot improve performance, since altering the aggregation strategy does not change the underlying error set.

4.1.5 Runtime and Memory Consumption

4.1.5.1 Runtime

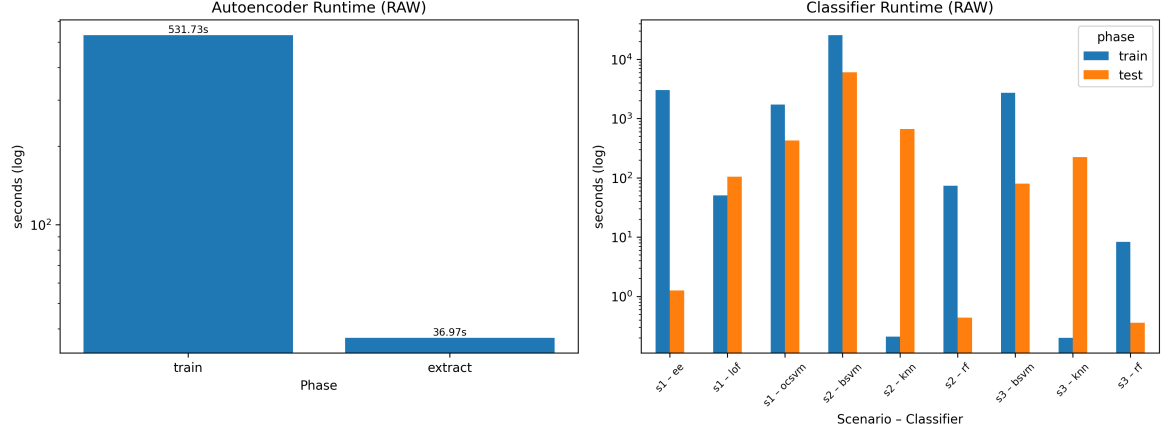


Figure 20: Runtime of Feature Extraction and Traditional ML Models — RAW

For the RAW representation (fig. 20), classifier runtimes vary drastically across models and scenarios. In general, training requires more time than testing; however, important exceptions exist.

Binary SVM exhibits the highest computational cost, particularly in Scenario 2, where both training and testing times are extremely large. Elliptic Envelope and OCSVM also incur substantial training costs in Scenario 1. In contrast, KNN has an almost negligible training time, as it merely stores the training data. Its prediction phase, however, is significantly more expensive because each test instance requires distance computations to many stored samples. This asymmetry is especially pronounced for RAW features.

Random Forest achieves comparatively small testing times, particularly in Scenarios 2 and 3, where predictions are performed in sub-second ranges. Overall, for RAW features the classifier choice strongly influences total runtime, with SVM-based models and KNN representing the main computational bottlenecks.

When using the RE15 representation (fig. 21), classifier runtimes decrease dramatically. Except for Elliptic Envelope, whose training time remains noticeably higher than the others, most classifiers operate within sub-second to approximately one-second ranges during training. Testing times are consistently small and relatively similar across models.

In this setting, the classifier stage is no longer the dominant cost factor. Instead, the primary computational overhead stems from the feature extraction step (autoencoder training), while the subsequent classification becomes comparatively lightweight.

Overall, the results indicate that RAW representations impose significant computational demands at the classifier level, whereas RE15 substantially reduces classifier runtime, shifting the computational burden primarily to feature extraction.

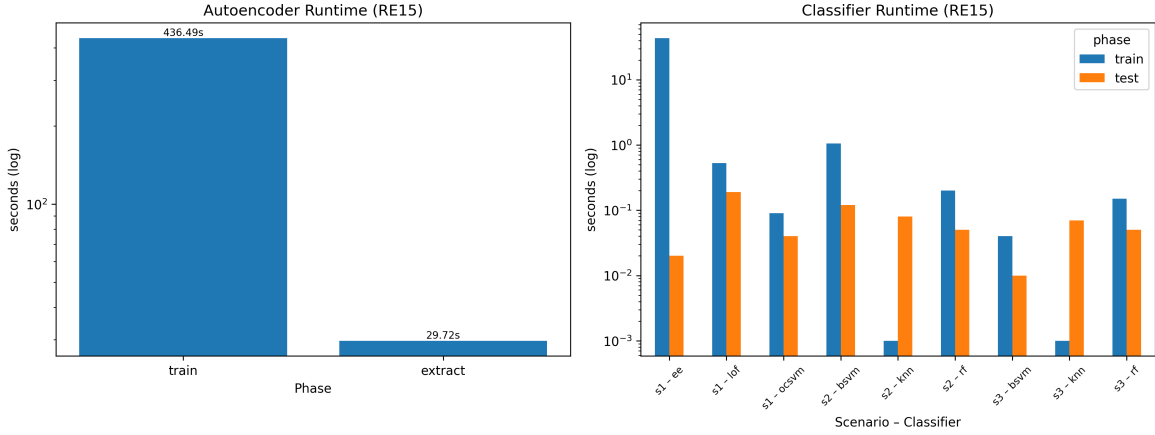


Figure 21: Runtime of Feature Extraction and Traditional ML Models — RE15

4.1.5.2 Memory Consumption

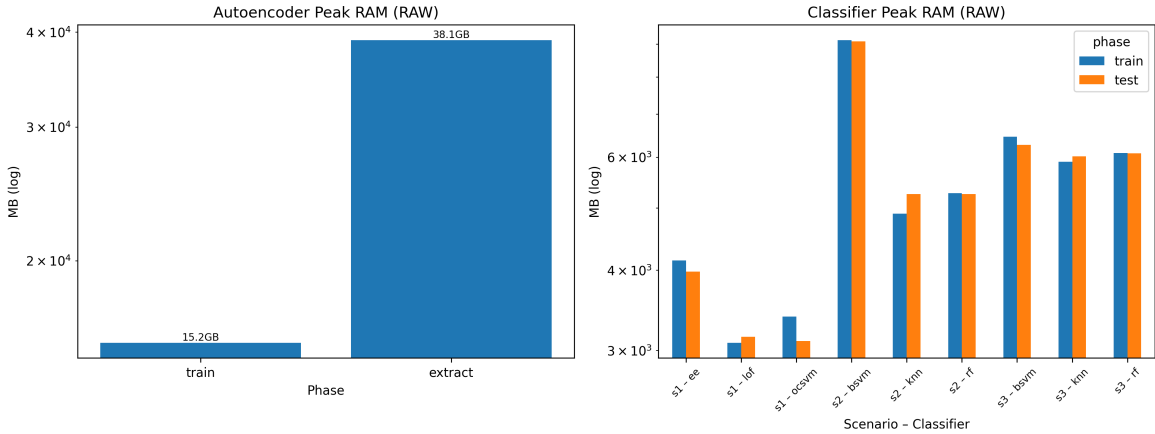


Figure 22: Memory Consumption — Feature Extraction and Traditional ML Models (RAW)

For the RAW representation (fig. 22), the feature extraction stage dominates overall memory usage. While autoencoder training already requires substantial memory, the encoding (feature extraction) phase reaches the highest peak RAM values within the entire pipeline. This indicates that processing and storing large byte-vector batches constitutes the primary memory bottleneck.

In contrast, classifier memory usage remains moderate relative to feature extraction. For RAW features, peak RAM consumption varies considerably across scenarios. Scenario 1 exhibits comparatively low memory usage, whereas Scenarios 2 and 3 require significantly more memory. The highest peak occurs for Binary SVM in Scenario 2, reflecting the cost of handling high-dimensional feature spaces with large training sets.

Across classifiers, training and testing phases show similar memory profiles, indicating that model size and stored data structures dominate memory requirements rather than temporary

computation during prediction.

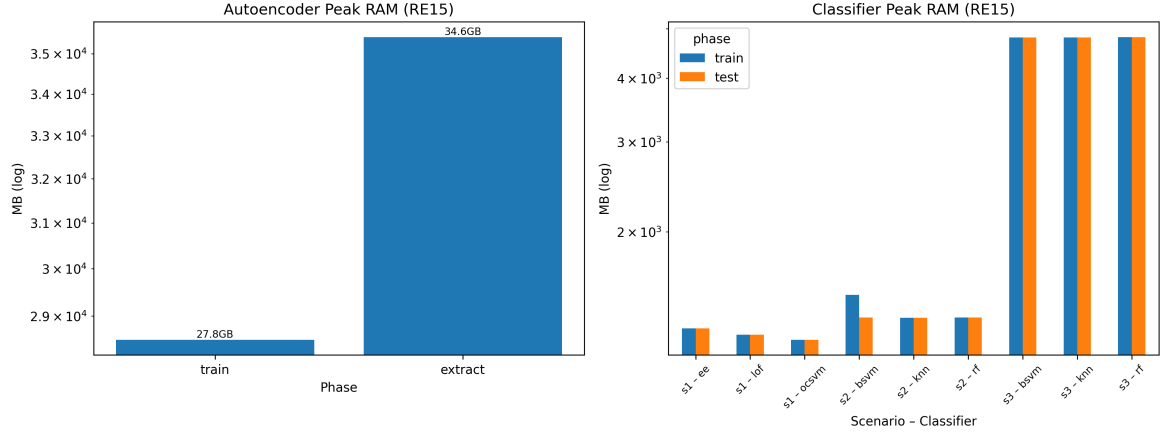


Figure 23: Memory Consumption — Feature Extraction and Traditional ML Models (RE15)

For the RE15 representation (fig. 23), overall memory consumption is reduced compared to RAW, particularly at the classifier stage. Nevertheless, the autoencoder remains the most memory-intensive component. Interestingly, autoencoder training requires more memory for RE15 than for RAW, suggesting that the representation structure influences internal activation storage.

Classifier memory usage under RE15 is substantially lower in Scenarios 1 and 2, but increases noticeably in Scenario 3. This scenario-dependent behavior indicates that dataset composition (e.g., class distribution and sample count) has a strong influence on memory allocation.

Overall, RAW exhibits lower memory in Scenario 1 but significantly higher memory in Scenarios 2 and 3, whereas RE15 maintains low memory usage in the first two scenarios but shows a marked increase in Scenario 3.

4.2 DL Models

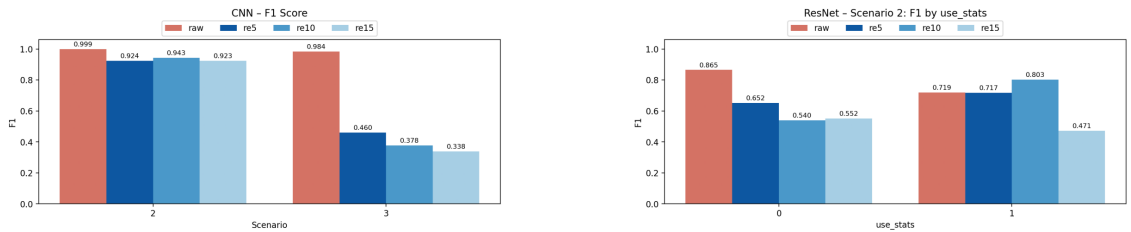


Figure 24: F1-scores of the CNN and ResNet classifiers across dataset representations

4.2.1 CNN Results

The left panel of fig. 24 summarizes the F1-scores of the CNN classifier for Scenarios 2 and 3 across the four dataset representations (RAW, RE5, RE10, RE15). All CNN models were trained with oversampling to mitigate class imbalance effects.

Across all dataset types, Scenario 2 achieves consistently strong performance. F1-scores exceed 0.9 for every representation, with near-perfect detection for the RAW dataset. This indicates that when sufficient labeled data and attack diversity are available, the CNN can effectively learn discriminative byte-level patterns.

Although the reverse-engineered representations perform slightly worse than RAW in Scenario 2, the differences among RE5, RE10, and RE15 remain relatively small. Among them, RE10 achieves the highest F1-score, suggesting that this representation offers a balanced trade-off between compression and preserved structure. Overall, the CNN appears capable of partially compensating for information loss in reduced representations when attack diversity is sufficient.

In contrast, Scenario 3 shows a substantial performance drop for all reverse-engineered datasets. Despite oversampling, F1-scores decrease markedly, indicating limited generalization capability. A likely explanation is the restricted diversity of attack types in this scenario. While oversampling increases the number of training instances, it does not introduce additional structural variation, which limits the model's ability to learn robust decision boundaries.

Notably, the RAW representation maintains strong performance even in Scenario 3. This suggests that full packet information provides rich structural cues that remain discriminative even under constrained attack configurations. In comparison, the reverse-engineered representations appear to be limited more by representational expressiveness than by class imbalance alone.

4.2.2 ResNet

The results obtained with the ResNet classifier in Scenario 2 show a clear impact of both reverse engineering and the inclusion of statistical features. When using raw packets without reverse engineering, the model achieved an F1-score of 0.865 without statistical features and 0.719 when statistical features were included. This indicates that raw packet information alone already provides strong discriminative signals for the classifier.

After applying reverse engineering, where only sequences of physical readings were used, the performance decreased. For $p = 5$, the F1-score dropped to 0.652, and further decreased to 0.540 for $p = 10$. A slight improvement was observed for $p = 15$, reaching 0.552, but the performance remained significantly below the raw packet baseline.

When statistical features were added to the ResNet input, their impact depended on the sequence length. For $p = 5$, the performance slightly improved to 0.717, suggesting that the additional statistical context helped compensate for the loss of structural packet information. For $p = 10$, the F1-score further increased to 0.803, representing the best result among the reverse-engineered inputs. However, for $p = 15$, the performance dropped significantly to

0.471, indicating that larger physical-reading sequences combined with statistical features may introduce noise or redundant information that negatively affects classification performance.

Overall, these results show that ResNet can effectively learn patterns from raw packet bytes, while its performance becomes more sensitive when only partial packet information is available.

Comparison CNN ResNet A comparison between CNN and ResNet shows noticeable differences in their robustness across scenarios. For raw packets, both models achieved high performance, with CNN slightly outperforming ResNet, reaching an F1-score close to 0.999, while ResNet achieved 0.865. This indicates that CNN is particularly effective at learning local byte-level patterns in raw packet structures.

However, when reverse engineering was applied and only partial packet data was used, both models experienced a decline in performance. The CNN showed a sharper drop in Scenario 3, whereas the ResNet demonstrated more stable behavior when statistical features were included. In particular, ResNet benefited from the additional statistical features in some configurations, reaching 0.803 for $p = 10$, which suggests that the deeper architecture and residual connections may help integrate heterogeneous input features more effectively.

In summary, CNN achieves the best performance when complete packet structures are available, while ResNet shows slightly better adaptability when additional statistical features are incorporated into the model.

4.3 Other Anomaly Detection Methods

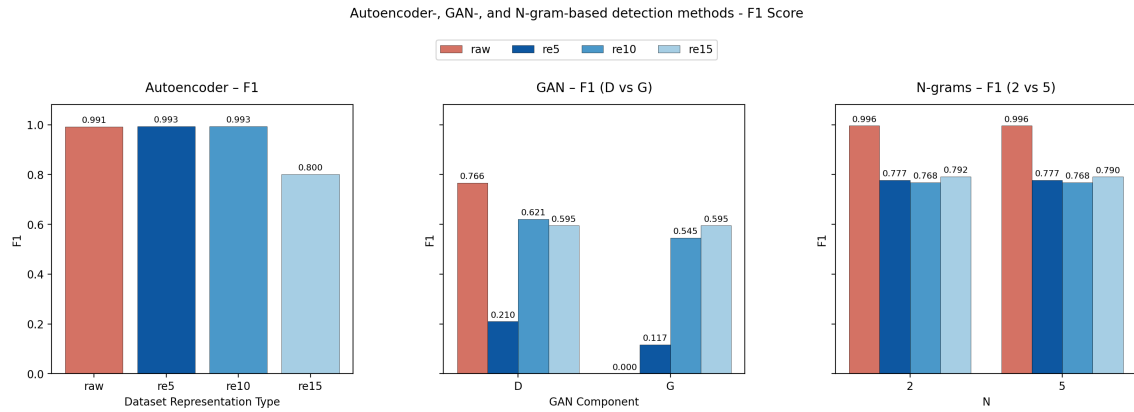


Figure 25: Autoencoder, GAN and N-Gram -based detection F1-Score

Autoencoder

The left panel of fig. 25 summarizes the F1-scores of the reconstruction-error-based SDA classifier across the different dataset representations (RAW, RE5, RE10, RE15).

For RAW, RE5, and RE10, the autoencoder achieves near-perfect classification performance, with F1-scores close to 1.0. Both precision and recall remain consistently high in these settings, indicating that the learned reconstruction threshold effectively separates benign and malicious traffic. Compared to RAW, the RE5 and RE10 representations even show slightly higher precision and F1-scores, suggesting that moderate reverse engineering does not degrade the discriminative power of the reconstruction-based method.

In contrast, performance drops dramatically for RE15. The F1-score decreases sharply, accompanied by both low recall and low precision. The reduced recall indicates that many attack instances are incorrectly classified as benign (false negatives), while the low precision reflects a substantial proportion of false positives among predicted attacks. This behavior suggests that the RE15 representation no longer preserves sufficient structure for reliable reconstruction-based anomaly detection.

Overall, the results demonstrate that the SDA-based approach is highly effective for RAW and moderately reduced representations (RE5, RE10), but becomes unstable when the reverse-engineered representation is further compressed.

GAN

As shown in the middle panel of figure 25, the performance of the GAN-based approach is highly unstable and significantly lower compared to the other evaluated methods. While the discriminator achieves better results than the generator-based detection, its performance remains inferior to that of the SDA and the N-gram models.

In particular, generator-based anomaly detection struggles considerably, especially when applied to raw packet data. We attribute this behavior to the inherent difficulty of generating valid raw network packets from random noise. Raw packets follow strict structural and syntactic constraints, including precise byte-level patterns and protocol-specific rules. Even minor deviations from these constraints can result in unrealistic or invalid packet representations.

Due to this high structural rigidity, the generator faces substantial challenges in learning an accurate representation of the underlying packet distribution. Consequently, the generated samples often fail to approximate realistic packet structures, leading to weaker anomaly detection performance.

N-Gram

The N-gram-based detection method demonstrates outstanding performance when applied to raw packet data, achieving an F1-score of approximately 0.996 as shown in the right panel of figure 25. This near-perfect performance indicates that raw network packets contain highly distinctive local byte-sequence patterns that can be effectively captured by N-gram representations. Since raw packets include both protocol headers and payload information, they preserve structural regularities and fixed byte-level dependencies that strongly differentiate normal and malicious traffic.

After applying reverse engineering and restricting the input to physical readings only, a noticeable decrease in performance can be observed. Although the F1-score drops compared to the raw packet scenario, it remains relatively stable across different values of the concatenation

parameter p . This stability suggests that increasing the number of concatenated physical readings does not significantly enhance or degrade the discriminating capability of the N-gram model.

The comparison between 2-gram and 5-gram configurations shows only marginal differences in performance. This indicates that extending the N-gram length does not substantially improve detection accuracy. In practice, shorter byte dependencies already capture most of the relevant structural information present in the packets. Larger N-grams may increase representational complexity, but they do not provide a significant additional benefit in this context.

The overall performance reduction observed after reverse engineering can be explained by the removal of protocol-related bytes. Raw packets contain strict and repetitive structural patterns defined by the underlying communication protocol. These regular patterns are highly suitable for modeling with N-grams, as they rely on local sequential dependencies. In contrast, physical readings are more variable and less structured at the byte level. As a result, the statistical regularities that N-grams exploit become weaker, leading to a reduced but still stable detection performance.

Comparison AE, GAN, N-Gram

Overall, the experimental results indicate that SDA-based approaches provide the most robust and consistent performance across all evaluated scenarios. In particular, the stacked denoising autoencoder (SDA) maintains stable detection accuracy not only on raw packet data but also after reverse engineering and data reduction. This robustness suggests that representation-learning methods are well-suited for modeling both protocol-level structures and higher-level behavioral patterns in ICS traffic, even when structural information is partially removed.

The N-gram-based approach also demonstrates strong performance, especially when applied to raw packet data. Its effectiveness can be attributed to its ability to capture local byte-level patterns and repetitive protocol structures. However, its performance degrades once the input is restricted to physical readings only, indicating that it relies heavily on the structural regularities present in complete packet data. While N-grams remain competitive, their dependence on raw packet structure limits their generalizability in scenarios where protocol information is unavailable or intentionally removed.

In contrast, GAN-based methods exhibit unstable and comparatively weak performance. Although the discriminator performs moderately better than the generator-based detection, the overall results suggest that GANs are not well-suited for reliable anomaly detection at the byte level in ICS environments. The strict structural constraints and deterministic nature of raw network packets make it difficult for the generator to learn a realistic data distribution from noise. Consequently, the generated samples often fail to approximate valid packet structures, which negatively impacts detection performance.

In summary, representation-learning approaches such as autoencoders appear to be the most reliable choice for anomaly detection in proprietary ICS traffic. N-grams offer strong performance when full structural information is available, whereas GAN-based models struggle to provide stable and trustworthy results in this highly structured and safety-critical domain.

4.4 Summary

This section summarizes the main observations across all evaluated approaches.

4.4.1 Overall Model Performance

Across all experiments, the strongest overall performance is achieved by the reconstruction-based SDA, the N-gram model, and the CNN classifier in Scenario 2. In this setting, F1-scores approach near-perfect classification, particularly for the RAW representation.

In contrast, GAN-based detection and traditional machine learning classifiers exhibit the weakest overall performance. While GAN models show moderate success in some configurations, their results remain substantially below those of autoencoder- and CNN-based approaches. Traditional ML classifiers are especially sensitive to the chosen representation and scenario and frequently suffer from significant performance degradation.

4.4.2 Influence of Dataset Representation

The RAW packet representation consistently yields the best results across nearly all model families. It preserves the full structural and contextual information of network traffic, enabling both deep learning and classical models to extract highly discriminative patterns.

The reverse-engineered representations (RE5, RE10, RE15) introduce progressive aggregation and information reduction. Among them, RE15—representing the strongest compression—produces the weakest overall performance. This indicates that aggressive abstraction of packet structure removes discriminative signals that are essential for reliable attack detection.

4.4.3 Impact of Reverse Engineering Across Model Families

The influence of reverse engineering differs substantially between model classes.

Deep learning architectures such as CNN, SDA, and ResNet show comparatively robust behavior. Although minor performance reductions occur for reduced representations, these models remain competitive, particularly when sufficient labeled data is available. This suggests that deep architectures can internally learn hierarchical feature representations that partially compensate for information loss.

GAN-based and N-gram models exhibit moderate performance degradation under reverse-engineered representations. While still functional, their discriminative capacity decreases more noticeably than for CNN or SDA.

The strongest negative impact is observed for traditional machine learning classifiers. These approaches rely explicitly on the extracted latent feature vectors and do not perform hierarchical representation learning. Consequently, they are more sensitive to the loss of structural information introduced by reverse engineering.

4.4.4 Impact of Evaluation Scenario

Scenario characteristics have a strong influence on performance across all models. Scenario 2 consistently achieves the highest F1-scores, indicating that models perform best when attack types are well represented during training.

Scenario 3 yields the lowest overall performance. The presence of unseen or highly constrained attack types significantly reduces generalization capability, even for strong deep learning models. This highlights that attack diversity during training plays a more decisive role than class imbalance alone.

5 Conclusion

5.1 Conclusion

This study project investigated whether anomaly detection can operate without explicit protocol knowledge while maintaining strong performance on reverse-engineered traffic representations.

The results show that protocol-agnostic detection is feasible, particularly when operating on raw packet data. Autoencoder-based detection, CNN models, and N-gram approaches achieve consistently strong performance, especially in Scenario 2 where sufficient attack diversity is available.

Reverse engineering reduces computational cost but introduces representation-dependent performance trade-offs. While deep learning models remain comparatively robust under moderate compression, traditional machine learning classifiers experience significant degradation. The most aggressively compressed representation (RE15) yields the weakest overall results, indicating that critical structural information is lost.

Scenario characteristics further influence performance: limited attack diversity (Scenario 3) in the training data consistently reduces generalization capability across all model families.

In summary, raw packet representations provide the most reliable detection performance, whereas reverse-engineered representations offer efficiency gains at the cost of reduced robustness. The optimal choice therefore depends on the trade-off between computational constraints and detection accuracy.

5.2 Limitations

Several limitations should be considered when interpreting the results of this study project.

First, a post-hoc analysis revealed structural characteristics of the dataset that may have influenced detection performance. In particular, the attacker IP address appears to remain constant across attack samples. Since the RAW representation includes full packet headers, including IP-level information, models trained on RAW data may have partially exploited this invariant identifier. In contrast, the reverse-engineered representations exclude such header information, which may contribute to the observed performance gap between RAW and RE-based datasets. Consequently, the absolute performance of RAW-based models should be interpreted with caution.

Second, attack labels were derived from the dataset folder structure rather than from packet-level ground-truth annotations. As a result, benign server responses that occurred within

attack sessions were also assigned the attack label. This introduces label noise and may have affected both training and evaluation, particularly for models sensitive to fine-grained traffic structure.

Finally, the evaluation was conducted on a specific dataset and predefined scenarios. The observed trends may vary under different network environments, attacker diversity, or alternative reverse-engineering strategies. Future work should incorporate datasets with greater attacker variability, stricter packet-level labeling, and validated process-value annotations to further assess robustness.

5.3 Future Work

The results of this study project open several avenues for further investigation.

The reverse-engineered representations rely on byte-level concatenation of extracted payload segments. Although aggregation controlled by the parameter p increases contextual information, it may obscure fine-grained temporal and structural dependencies within individual packets. Future work could explore sequence-aware modeling approaches that preserve packet boundaries or explicitly encode temporal order instead of relying solely on concatenation.

The traditional ML classifiers operate on latent features extracted by the stacked denoising autoencoder. While this compact representation proved effective in certain settings, alternative feature extraction strategies may further improve performance. For example, statistical traffic descriptors, protocol-agnostic sequence embeddings, or transformer-based encoders could provide richer and more expressive representations for classical machine learning models.

Moreover, the evaluation was conducted on a dataset with specific structural characteristics, including limited attacker diversity and coarse-grained labeling. Future studies should therefore consider more heterogeneous datasets with diversified attacker behavior, strict packet-level annotations, and validated process-value ground truth. Such validation would allow for a more robust assessment of generalization performance under realistic industrial conditions.

Abbreviations and Acronyms

ICS	Industrial Control Systems
OT	Operational Technology
OCC	One Class Classifier
IT	Information Technology
NTP	Network Time Protocol
IP	Internet Protocol
NBNS	NetBIOS Name Service
mDNS	Multicast Domain Name Service
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
HTTP	Hyper Text Transfer Protocol
HTTPS	Hyper Text Transfer Protocol Secure
PLCs	Programmable Logic Controllers
SCADA	Supervisory Control and Data Acquisition
GANs	Generative adversarial networks
ML	Machine Learning
DL	Deep Learning
S7comm	Siemens S7 communication
pcap	Packet Capture
RE	Reverse-Engineered
MSE	Mean Squared Error
SVM	Support vector machine
BSVM	Binary Support vector machine
OCSVM	One Class-Support vector machine
LOF	Local outlier factor
KNN	k Nearest Neighbor
EE	Elliptic Envelope
CNN	Convolutional Neural Network
ResNet	Residual Network

SDA	Stack Denoising Autoencoder
RPB	Ratio of padded bytes
ReLU	Rectified linear unit
FMR	False Matching Rate
FNMR	False Non-Matching Rate

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